

Implementation of Long Short-Term Memory (LSTM) Models for the Prediction of Vegetation Indices (NDVI, NDMI) from Time Series of Meteorological Variables for Lombardy Region, Italy'

Abstract

Accurate prediction and gap-filling of vegetation indices are fundamental requirements for operational ecological monitoring, precision agriculture, and regional water resource management. In topographically complex regions like the Lombardy region of northern Italy, reconstructing continuous vegetation dynamics from point-based sensor networks remains a critical challenge. This study implements, evaluates, and comparatively analyses four Long Short-Term Memory (LSTM)-based deep learning architectures—a stacked LSTM, a Bidirectional LSTM (BiLSTM), a Smoothed-BiLSTM incorporating Blackman finite impulse response (FIR) pre-filtering, and a novel Attention-BiLSTM—for the simultaneous prediction of the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Moisture Index (NDMI). Multivariate predictor sets were assembled for 5 unique and distinct sensor locations (PIDs) drawn from a broader regional network of 3,955 sensors across Lombardy's heterogeneous landscape, utilizing meteorological variables (*Temperature, Precipitation, Relative Humidity*), static topographic descriptors (*Elevation, Slope, Northerness, Northerness_Slope*), land-cover classifications (*Fire_Fuel_Class*), and historical vegetation index values. Models were trained on 2022 data and evaluated on an independently withheld 2023 test set using a comprehensive 12-metric framework, including RMSE, R^2 , Kling-Gupta Efficiency (KGE), and Nash-Sutcliffe Efficiency (NSE), alongside seasonal stratification and pairwise Wilcoxon signed-rank tests. The Smoothed-BiLSTM achieved superior overall NDVI prediction performance (RMSE = 0.0445, R^2 = 0.9853, KGE = 0.9800), closely followed by the baseline LSTM (RMSE = 0.0483, R^2 = 0.9827, KGE = 0.9775). For NDMI prediction, the baseline LSTM ranked first (RMSE = 0.0595, R^2 = 0.9126, KGE = 0.8314), highlighting the differing temporal dynamics of canopy moisture compared to vegetation greenness. Pairwise Wilcoxon signed-rank tests confirmed that performance differences between the LSTM and Smoothed-BiLSTM for NDVI were statistically non-significant (p = 0.368), while other architectural comparisons were highly significant (p < 0.001). Furthermore, a timestep ablation study identified four recent observations (= 4 days) as the practical minimum temporal context window required for reliable gap-filling. These results advance the state of the art in vegetation index reconstruction for alpine-Mediterranean landscapes and provide directly transferable guidance for operational sensor-network data systems in topographically complex regions.

Keywords: NDVI; NDMI; LSTM; Bidirectional LSTM; Attention mechanism; gap-filling; Lombardy; time series prediction; Blackman FIR filter; sensor networks.

1. Introduction

Vegetation indices derived from continuous environmental monitoring constitute the observational backbone of contemporary ecological assessment, precision agriculture, and regional resource management. Among these, the Normalized Difference Vegetation Index (NDVI) remains the most widely deployed metric for characterising vegetation greenness, biomass, and phenological state (Rouse et al., 1974), while the Normalized Difference Moisture Index (NDMI) provides complementary sensitivity to canopy water content and moisture stress, which NDVI alone cannot resolve (Gao, 1996; Wilson and Sader, 2002). The temporal dynamics of both indices encode the integrated response of vegetation to climate forcing, land management, and landscape structure. Their continuous monitoring is foundational to applications ranging from crop yield estimation and drought early warning to fire risk assessment and carbon flux modelling (Farbo et al., 2024; Guo et al., 2024; Chuvieco et al., 2010).

While the deployment of dense, point-based sensor networks has substantially expanded observational capacity for vegetation monitoring at local and regional scales, reconstructing complete, gap-free time series from these networks remains a critical challenge. In topographically complex regions like the Lombardy region of northern Italy, sensor data frequently suffer from missing values due to transmission failures, instrument downtime, power outages, or intermittent maintenance. This creates both a scientific challenge—incomplete time series impede phenological analysis and model calibration—and an operational challenge for real-time precision agriculture and environmental monitoring systems that depend on reliable, continuous vegetation index products.

The problem of vegetation index gap-filling—reconstructing missing observations from available meteorological, topographic, and historical spectral data—has been addressed through a series of methodological frameworks. Classical interpolation methods, including linear and cubic spline interpolation, and statistical smoothing approaches such as the Savitzky–Golay filter and Fourier decomposition, address short gaps effectively but lack the capacity to extrapolate vegetation dynamics over extended missing periods or to leverage heterogeneous predictor information (Michishita et al., 2014; Hird and McDermid, 2009). Physically-based process models can simulate vegetation dynamics from meteorological inputs but require extensive calibration and fail to generalise across the diverse vegetation communities and topographic contexts present in a complex region such as Lombardy. Machine learning approaches, including Random Forest and Support Vector Machine regression, have demonstrated skill in time series prediction tasks but are not natively designed to capture the deep temporal autocorrelation and sequential dependencies inherent in time series data (Farbo et al., 2024).

Deep learning architectures based on Recurrent Neural Networks (RNNs)—and specifically Long Short-Term Memory (LSTM) networks—have emerged as the state-of-the-art methodology for vegetation index time series prediction. This is due to their capacity to learn long-range temporal dependencies, handle multivariate inputs, and generalise across diverse environmental conditions (Hochreiter and Schmidhuber, 1997; Guo et al., 2024; Farbo et al., 2024). Guo et al. (2024) demonstrated that a multivariate LSTM incorporating climate elements outperformed conventional time series models for monthly NDVI prediction in the Yellow River Basin. Farbo et al. (2024) developed a double-layer Bidirectional LSTM (BiLSTM) for field-level NDVI forecasting of corn crops in Piemonte (NW Italy). Furthermore, Dey et al. (2024) proposed a Smoothed-LSTM architecture applying Blackman finite impulse response (FIR) pre-filtering to NDVI time series, demonstrating that input preprocessing significantly reduces outlier sensitivity and improves the temporal smoothness of LSTM predictions. While these approaches have predominantly been applied to satellite-derived data, their methodological framework is highly transferable to, and critically needed for, high-frequency point-based sensor network data in complex terrain.

Despite these advances, critical research gaps remain that motivate the present study. First, existing LSTM-based vegetation index prediction works have focused predominantly on a single spectral index (NDVI), while canopy moisture dynamics captured by NDMI have received negligible attention in the deep learning gap-filling literature. Given the growing importance of moisture stress indicators for drought monitoring and fire risk assessment, simultaneous prediction of both NDVI and NDMI from shared multivariate inputs represents an important and under-explored problem. Second, most existing studies have been conducted over relatively homogeneous landscapes, and the transferability of architectural findings to topographically and ecologically complex alpine–sub-Alpine landscapes using dense sensor networks has not been established. Third, while several architectural variants (standard LSTM, BiLSTM, attention-augmented LSTM) have been proposed independently in the literature, systematic comparative evaluation under identical data conditions for vegetation index reconstruction tasks remains scarce, limiting the ability to draw generalised architectural recommendations.

The present study addresses these gaps through a rigorous, multi-architecture comparative evaluation of four LSTM-based models for simultaneous NDVI and NDMI gap-filling over the Lombardy region. The analysis utilises a representative subset of 5 distinct sensor locations (PIDs) drawn from a broader regional network of 3,955 sensors, spanning the years 2022 (training) and 2023 (held-out testing). The study area—encompassing a pronounced altitudinal gradient from the Po Plain to the Alpine peaks, and featuring intensive agriculture, mixed forest, Alpine meadow, and peri-urban landscapes—provides an exceptionally demanding evaluation context for model generalisation. The four architectures evaluated—a stacked LSTM, a Bidirectional LSTM, a Smoothed-BiLSTM incorporating Blackman FIR preprocessing, and a novel Attention-BiLSTM—represent a systematic progression from established baselines to recently proposed innovations, enabling both direct comparison and identification of the specific architectural features that confer performance advantages.

The specific objectives of this study are:

- (1) to implement and comparatively evaluate four LSTM-based architectures for NDVI and NDMI gap-filling using a comprehensive 21-variable multivariate predictor set. This set integrates meteorological variables (Temperatura, Precipitazione, Umidita_Relativa), static topographic descriptors (Elevazione, Slope, Northerness, Northerness_Slope), cyclic temporal encodings (sin_DoY, cos_DoY), autoregressive lag features (NDVI_lag1, NDVI_lag7, NDMI_lag1, NDMI_lag7, Temp_lag1), rolling statistics (Precip_sum7, NDVI_roll7, NDMI_roll7), and Blackman FIR-smoothed spectral inputs (NDVI_smooth, NDMI_smooth);
- (2) to assess seasonal, phenological, and terrain-stratified model performance, with particular attention to the differential behaviour of architectures across land cover types, slope classes, and aspect orientations;
- (3) to characterise the minimum temporal context window required for reliable gap-filling through a systematic timestep ablation study; and
- (4) to provide statistically rigorous model comparisons using pairwise Wilcoxon signed-rank tests across the full residual distributions.

The remainder of the study is structured as follows. Section 2 describes the study area, the sensor network dataset, and the derived predictors. Section 3 details the experimental design, model architectures, training procedures, and evaluation framework. Section 4 presents and discusses results, including training convergence, multi-metric test set performance, residual diagnostics, seasonal and topographic stratification, and timestep ablation. Section 5 synthesises the main conclusions and outlines directions for future research.

2. Materials

2.1 Study Area

The study area encompasses the Lombardy region (Lombardia) of northern Italy, located between approximately 44°41'N–46°38'N latitude and 8°31'E–11°08'E longitude, covering a total surface area of approximately 23,863 km². Lombardy represents one of the most ecologically and agriculturally complex regions in Italy, spanning a pronounced altitudinal gradient from the Po Plain in the south (ranging from approximately 50 m a.s.l.) through the Prealps to the Alpine peaks of the north, which exceed 4,000 m a.s.l. (Ghilardi et al., 2023). This topographic diversity generates a corresponding diversity in vegetation cover, land use, and microclimatic conditions, making the region particularly well suited for the evaluation of predictive models for vegetation indices.

The region encompasses a variety of land cover types, including intensive agricultural cropland concentrated in the Po Plain (principally maize, rice, and soybean), mixed broadleaf and conifer forests across the Pre-Alpine belt, Alpine meadows and shrubland at higher elevations, and peri-urban and urban areas around the metropolitan core of Milan. This heterogeneity introduces significant spatial variability in NDVI and NDMI dynamics across different sensor locations, presenting a rigorous test case for LSTM-based time-series prediction. The climate of Lombardy is highly variable across elevation zones: the lowland Po Plain is characterised by a humid subtropical Köppen classification (Cfa), with hot, humid summers and cold, foggy winters; the Pre-Alpine belt transitions toward temperate oceanic conditions (Cfb); and the high Alpine areas exhibit cold continental to tundra climates (Dfc and ET), with snow cover persisting from October to May (Beck et al., 2018). Mean annual precipitation across the region ranges from approximately 700 mm in the lowland eastern sectors to over 1,800 mm in the orographic precipitation maxima of the lake districts and Alpine foothills. Mean annual temperature ranges from approximately 13°C in the Po Plain to below 0°C at high Alpine elevations.

Given the operational realities of environmental monitoring in such a diverse and demanding landscape, dense point-based sensor networks are essential for capturing localised vegetation and microclimatic dynamics. However, continuous data acquisition from these ground-based networks is frequently disrupted by harsh environmental conditions—including extreme winter cold, heavy snowfall, and the logistical challenges of remote deployments—leading to transmission failures, instrument downtime, and power outages. This creates both a scientific challenge, as incomplete time series impede phenological analysis and model calibration, and an operational challenge for real-time precision agriculture and environmental monitoring systems that depend on reliable, gap-free vegetation index products. Consequently, Lombardy serves as an exceptionally demanding and operationally relevant case study for evaluating deep learning-based gap-filling and predictive models for sensor-derived vegetation indices.

2.2 Satellite Data and Vegetation Indices

The dataset comprises time-series observations from a regional environmental sensor network consisting of 3,955 distinct sensor locations, each assigned a unique identifier (PID). The available data spans the years 2022 and 2023. For the purpose of rigorous deep learning pipeline evaluation and computational feasibility, a representative subset of 5 PIDs was explicitly selected for model training and testing, ensuring that the full temporal dynamics of the 2022–2023 period were captured while maintaining ecological representation across the network.

The Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Moisture Index (NDMI) serve as the primary target variables for this study. NDVI characterises vegetation greenness, biomass, and phenological state, while NDMI provides complementary sensitivity to canopy water content

and moisture stress. Unlike satellite-derived indices, these values are derived directly from the sensor network's proximal or spectral measurements, providing high-frequency, ground-truth-aligned observations. Missing observations in the time series were retained as artificial gaps to serve as the target for the LSTM gap-filling and prediction models, simulating real-world sensor dropout scenarios.

The Normalized Difference Vegetation Index (NDVI) is computed using:

$$NDVI = \frac{(B8 - B4)}{(B8 + B4)}$$

The Normalized Difference Moisture Index (NDMI) is computed using:

$$NDMI = \frac{B8 - B11}{B8 + B11}$$

A total of 5 PIDs were sampled across the Lombardy region using a stratified random sampling design, with strata defined by the four fuel or land-cover classes present in the dataset (Fire_Fuel_Class column), the three topographic elevation classes (below 500 m, 500–1000 m, and above 1000 m), and four slope classes. This stratified sampling guaranteed that all major vegetation and topographic types were represented in both training and test subsets, in contrast to purely random pixel selection which risks underrepresenting rare but ecologically important classes (as previously discussed in Section 1). Each sampled pixel was assigned a unique pixel identifier (PID), and all meteorological and ancillary variables were extracted at the corresponding spatial location.

2.3 Meteorological Data

Local meteorological forcing data are provided as integral, co-located components of the sensor network dataset for each PID. Unlike studies that rely on spatially interpolated gridded products, these direct measurements eliminate the spatial interpolation errors typically associated with distant weather station networks or coarse reanalysis grids, ensuring high fidelity between the meteorological drivers and the local vegetation response (Guo et al., 2024; Farbo et al., 2024). To capture the immediate environmental drivers of vegetation dynamics, the following daily meteorological variables were included in the predictor set (using the exact column nomenclature from the dataset):

1. **Temperature** (*Temperatura*, °C): Mean daily air temperature.
2. **Precipitation** (*Precipitazione*, mm): Daily cumulative precipitation.
3. **Relative Humidity** (*Umidita_Relativa*, %): Daily average relative humidity.

These variables provide the model with direct, high-frequency atmospheric context, which Guo et al. (2024) demonstrated are essential climate elements for multivariate LSTM models predicting vegetation index dynamics.

2.4 Topographic Data

Static topographic descriptors were associated with each PID to allow the model to learn landscape-level controls on vegetation index dynamics, which is critical in topographically complex terrain. The following static variables were included:

Elevation (*Elevazione*, m a.s.l.): Raw elevation of the sensor location.

Slope (Slope, degrees): Terrain gradient, computed using standard finite-difference algorithms (Horn, 1981), capturing drainage and soil stability conditions.

Northernness (Northernness, dimensionless): Computed as $\cos(\text{aspect})$, ranging from -1 (south-facing) to $+1$ (north-facing), capturing aspect effects on solar radiation receipt and evapotranspiration (Beers et al., 1966).

Northernness $\times$ Slope (Northernness_Slope): An interaction term capturing the combined effect of slope gradient and aspect orientation, which modulates both available solar energy and local microclimates.

These topographic variables are static in time and serve as conditioning predictors, consistent with approaches used to model topographic controls on vegetation in complex terrain in the Italian Alps (Ghilardi et al., 2023).

2.5 Fuel and Land Cover Classification

Land cover and vegetation fuel type were derived from the Corine Land Cover (CLC) 2018 dataset, reclassified into four fuel classes consistent with the European Forest Fire Information System (EFFIS) classification: (1) grassland and natural meadows, (2) transitional woodland-shrub and shrubland, (3) broadleaf and mixed forest, and (4) conifer forest and Alpine vegetation. This classification was stored as a categorical column (*Fire_Fuel_Class*) in the dataset and used for stratified pixel sampling, fuel-class performance analysis (Comment 1), and colour-coded visualisation. The mapping between CLC classes and EFFIS fuel types followed the methodology described by San-Miguel-Ayanz et al. (2012).

2.6 Derived Temporal Features

Following the approach established in multivariate LSTM frameworks for time-series forecasting (Guo et al., 2024), several temporal features were engineered from the raw sensor data to provide the model with richer representations of phenological and seasonal dynamics. All derived features were computed per PID, sorted chronologically by Date, to prevent information leakage. The complete feature set comprises exactly 21 predictors-

Base Spectral & Meteorological Variables: NDVI, NDMI, Temperatura, Precipitazione, Umidita_Relativa.

Static Topographic Descriptors: Elevazione, Slope, Northernness, Northernness_Slope.

Cyclic day-of-year encoding: To represent the seasonal position of each observation without introducing a discontinuity at the year boundary (day 365 to day 1), the day of year (DoY) was encoded using sine and cosine transformations:

$$\sin_DoY = \sin\left(\frac{2\pi \cdot DoY}{365}\right), \cos_DoY = \cos\left(\frac{2\pi \cdot DoY}{365}\right)$$

Base Spectral & Meteorological Variables: NDVI, NDMI, Temperatura, Precipitazione, Umidita_Relativa.

Static Topographic Descriptors: Elevazione, Slope, Northernness, Northernness_Slope.

Lag Features: Autoregressive predictors were derived to provide the model with explicit historical context beyond the recurrent memory encoded in the LSTM hidden state:

NDVI_lag1, NDVI_lag7: NDVI values at 1 and 7 time steps prior.

NDMI_lag1, NDMI_lag7: NDMI values at 1 and 7 time steps prior.

Temp_lag1: Air temperature at 1 time step prior.

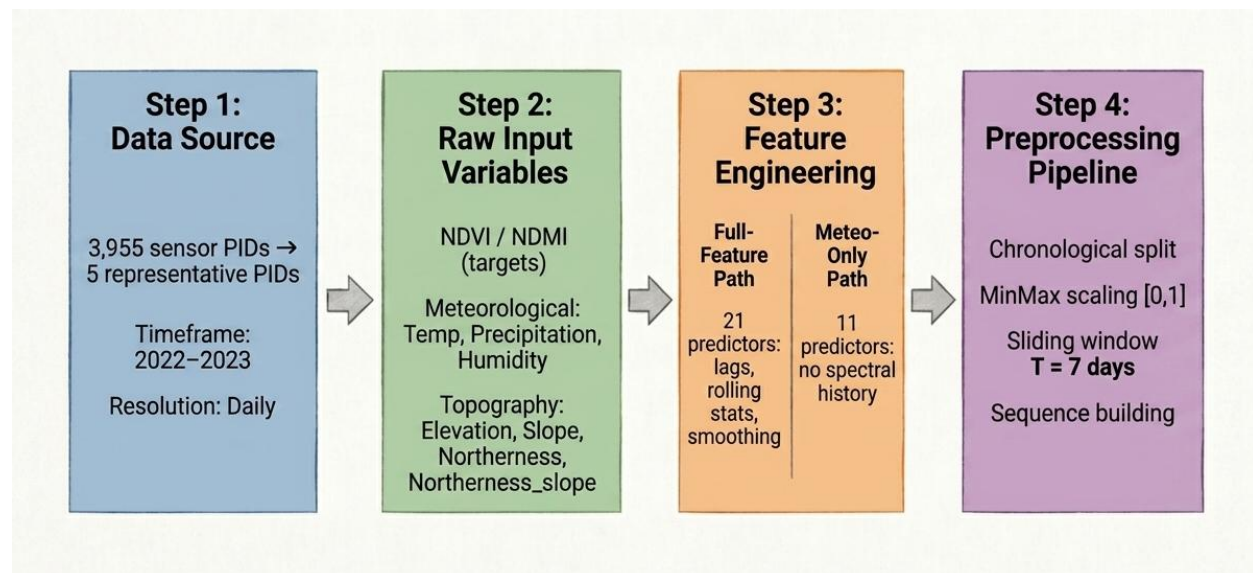
Rolling Statistics: Seven-step rolling aggregates were computed to capture medium-term meteorological forcing and vegetation trends:

Precip_sum7: Cumulative 7-step precipitation sum.

NDVI_roll7, NDMI_roll7: 7-step rolling mean of NDVI and NDMI.

Blackman FIR Smoothing: To reduce high-frequency noise in the NDVI and NDMI time series, a zero-phase Blackman finite impulse response (FIR) low-pass filter (order 31, normalised cutoff frequency 0.1) was applied per PID, producing NDVI_smooth and NDMI_smooth as additional predictor columns. As demonstrated by Dey et al. (2024), applying Blackman FIR pre-filtering to vegetation index time series significantly reduces outlier sensitivity and improves the temporal smoothness of LSTM predictions. Furthermore, as noted by Michishita et al. (2014), Blackman-windowed FIR filters outperform alternatives like Savitzky–Golay or simple rolling means in preserving phenological inflection points while suppressing sensor noise.

This comprehensive 21-variable feature set was normalised using MinMax scaling (fitted exclusively on the 2022 training data to prevent data leakage) and structured into sliding-window sequences (timesteps = 7, horizon = 1) for input into the deep learning architectures.



A- Input Data Preparation

3. Methods

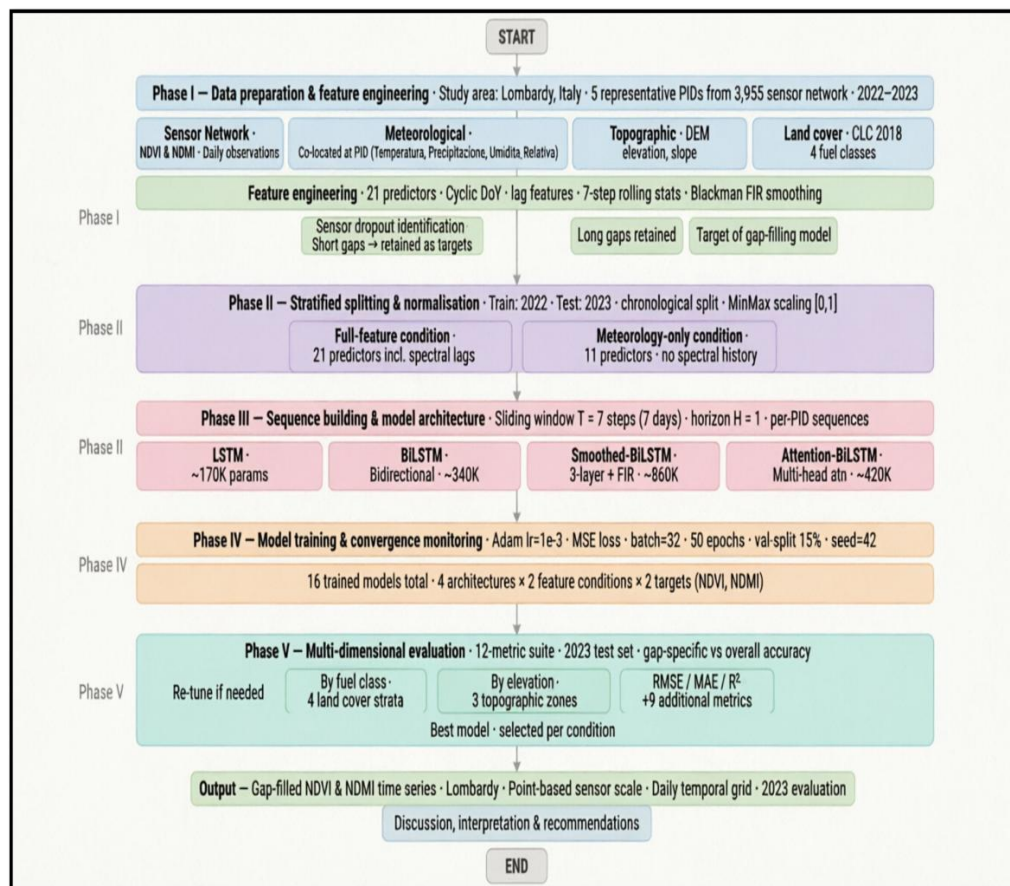
3.1 Experimental Design Overview

The methodological workflow comprised five sequential phases: (i) data preparation and feature engineering, (ii) stratified train–test splitting and normalisation, (iii) sequence building and model architecture design, (iv) model training and convergence monitoring, and (v) evaluation across multiple stratification dimensions. The workflow is summarised schematically in Figure 1 (conceptual representation). Inspired by the hierarchical modular pipeline approach of Farbo et al. (2024), each phase

was implemented as an independent Python module to allow partial re-runs without repeating computationally expensive upstream steps.

Two parallel model conditions were implemented and evaluated for both NDVI and NDMI prediction targets:

- **Full-feature models:** all 21 predictors including spectral index lags and smoothed variants, representing the standard gap-filling scenario where surrounding valid observations are available.
- **Meteorology-only models:** a reduced predictor set of 11 variables comprising only meteorological forcing and static topographic descriptors, explicitly excluding all NDVI/NDMI-derived features, representing the more challenging scenario where no surrounding valid spectral observations are available — analogous to the meteo-only ablation of Guo et al. (2024).



B- Methodological Flowchart

Both conditions were trained under four neural network architectures (LSTM, Bidirectional LSTM, Smoothed-BiLSTM, and Attention-BiLSTM), yielding a total of 16 trained models overall.

- **Full-Feature:** 4 architectures × 2 targets (NDVI, NDMI) = **8 models**
- **Meteo-Only:** 4 architectures × 2 targets (NDVI, NDMI) = **8 models**
- **Total = 16 Models**

3.2 Data Preparation and Preprocessing

3.2.1 Gap Identification and Signal Smoothing

The raw sensor network time series inherently contain missing observations due to transmission failures, instrument downtime, or power outages. Rather than interpolating these gaps prior to modelling, missing values were retained to simulate real-world sensor dropout scenarios, serving as the explicit targets for the gap-filling models.

To mitigate high-frequency sensor noise and improve the temporal coherence of the input signals, a zero-phase Blackman finite impulse response (FIR) low-pass filter was applied independently to the NDVI and NDMI time series for each PID. Following the methodology of Dey et al. (2024), the filter was configured with an order of 31 and a normalised cutoff frequency of 0.1. This preprocessing step generates the NDVI_smooth and NDMI_smooth features, which preserve phenological inflection points while suppressing transient atmospheric or instrumental noise more effectively than standard rolling means or Savitzky–Golay filters.

3.2.2 Train–Test Splitting

To prevent temporal data leakage and accurately replicate operational deployment conditions, a strict chronological split was applied. Data from the year 2022 were exclusively used for model training and validation, while data from the year 2023 served as the completely held-out test set. This approach is preferred over random splitting for ecological time series, as random splits artificially inflate performance metrics by allowing models to interpolate between temporally adjacent training and testing observations (Roberts et al., 2017).

3.2.3 Feature Scaling

All numerical predictors were normalised to the $[0, 1]$ range using MinMax scaling fitted exclusively on training data and subsequently applied to the test set, consistent with standard practice in deep learning for time series (Djordjević et al., 2022; Guo et al., 2024). Separate scalers were maintained for the full-feature and meteorology-only feature spaces, and for each prediction target (NDVI, NDMI), yielding four scaler objects in total. For the meteorology-only condition, where the target variable is not included in the input feature space, an additional univariate MinMax scaler was fitted for each target variable independently, allowing proper inverse transformation of model outputs for metric computation in original physical units.

3.3 Sequence Construction

3.3.1 Sliding Window Approach

Input sequences were constructed using a sliding window approach with a lookback window of $T = 7$ time steps and a prediction horizon of $H = 1$ time step. Each input sample $X_i \in \mathbb{R}^{T \times F}$ contains the $F = 21$ features observed over the preceding 7 time steps, and the corresponding label y_i is the target variable (NDVI or NDMI) at the immediately following time step. This one-step-ahead formulation is the canonical gap-filling setting, enabling iterative multi-step gap imputation by applying the model sequentially over a missing period.

Sequences were constructed independently per PID to prevent temporal context from bleeding across sensor boundaries. PIDs with fewer than $T + 1 = 8$ valid observations after preprocessing were excluded from sequence construction. The resulting arrays were saved as NumPy binary files to ensure efficient loading during model training.

3.3.2 Meteorology-Only Sequence Construction

For the meteorology-only condition, an identical sliding window approach was applied over the reduced 11-column feature matrix. The target variable (NDVI or NDMI) was excluded from the input features but retained as the prediction target. A separate univariate scaler was fitted on the training target values to normalise the output, and its parameters were saved for inverse transformation at evaluation time. This design directly operationalises Comment 2 of the project specification — training and evaluating models in a condition where no spectral index history is available, thereby quantifying the marginal contribution of spectral predictors to gap-filling skill relative to meteorological forcing alone.

3.4 Model Architectures

Four neural network architectures of increasing complexity were implemented, trained, and evaluated for each target variable and feature condition. All models were implemented in TensorFlow 2.14 (Keras API), using the Adam optimiser (Kingma and Ba, 2015) with a learning rate of 1×10^{-3} , Mean Squared Error (MSE) loss, and a mini-batch size of 32. Training proceeded for a maximum of 50 epochs with validation monitoring using a 15% split of the training data (VAL_SPLIT = 0.15). All models were initialised with a fixed random seed (SEED = 42) for reproducibility.

3.4.1 LSTM (Baseline)

The baseline model comprised two stacked LSTM layers (Hochreiter and Schmidhuber, 1997) with 128 and 64 units respectively, each followed by a 20% dropout layer for regularisation (Srivastava et al., 2014). Batch normalisation was applied after the second LSTM layer. A dense hidden layer with 32 units and ReLU activation preceded the single linear output unit. The two-layer stacked architecture was adopted following Guo et al. (2024), who demonstrated that a three-layer multivariate LSTM outperforms both shallower and deeper configurations for monthly NDVI prediction in the Yellow River Basin, with the two-layer variant providing a competitive baseline. The total parameter count is approximately 170,000.

Formally, for each time step t , the LSTM cell state C_t and hidden state h_t are updated as:

$$\begin{aligned} f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \\ i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\ \tilde{C}_t &= \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \\ C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \\ o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \\ h_t &= o_t \odot \tanh(C_t) \end{aligned}$$

where σ denotes the sigmoid activation function, $\odot$ element-wise multiplication, and W, b the learnable weight matrices and bias vectors of the forget (*f*), input (*i*), candidate cell ($\tilde{c}$), and output (*o*) gates (Greff et al., 2017).

3.4.2 Bidirectional LSTM (BiLSTM)

The BiLSTM model processes the input sequence in both forward and backward temporal directions simultaneously, concatenating the hidden states from both passes before the output layer (Graves et al.,

2005). This architecture allows the model to exploit future context within the observed window, which is beneficial for gap-filling (as opposed to forecasting) since observations both before and after the gap are typically available. The architecture comprised two stacked bidirectional LSTM layers with 128 and 64 units per direction, respectively, yielding approximately 340,000 parameters. The same dropout, batch normalisation, and dense output structure as the baseline LSTM was applied. The BiLSTM formulation is:

$$\begin{aligned}\vec{h}_t &= \vec{LSTM}(x_t, \vec{h}_{(t-1)}) \\ \overleftarrow{h}_t &= \overleftarrow{LSTM}(x_t, \overleftarrow{h}_{(t+1)}) \\ h_t &= [\vec{h}_t; \overleftarrow{h}_t]\end{aligned}$$

consistent with the formulation described by de Castro et al. (2020) and Farbo et al. (2024), who employed a 2-layer BiLSTM as their primary architecture for NDVI forecasting over corn fields in Piemonte.

3.4.3 Smoothed-BiLSTM

The Smoothed-BiLSTM extended the BiLSTM with a third bidirectional layer (units: 180, 80, 40 per direction across successive layers), explicitly designed to leverage the Blackman FIR-smoothed spectral features (NDVI_smooth, NDMI_smooth) as additional inputs, following the Smoothed-LSTM concept of Dey et al. (2024) and upgraded to a bidirectional formulation. By providing both raw and smoothed spectral inputs, the model can learn to exploit low-frequency phenological trends encoded in the smoothed series while retaining sensitivity to high-frequency weather-driven variability in the raw series. The three-layer design was motivated by the finding of Guo et al. (2024) that progressively deeper LSTM networks better capture hierarchical temporal features at multiple timescales, from short-term weather responses (first layer) through seasonal phenological cycles (second layer) to interannual trends (third layer). The approximate parameter count is 860,000, making this the most expressive architecture in the comparison set.

3.4.4 Attention-BiLSTM

The Attention-BiLSTM is the novel architecture proposed in this study, combining a bidirectional LSTM encoder with a multi-head self-attention mechanism. The BiLSTM encoder (128 units per direction) first produces a sequence of contextualised hidden states $\mathbf{H} \in \mathbb{R}^{T \times 256}$. Multi-head self-attention with $n_heads = 4$ and key dimension $d_k = 64$ is then applied over this sequence:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

A residual connection followed by layer normalisation (He et al., 2016; Ba et al., 2016) integrates the attention output with the encoder representations:

$$\mathbf{Z} = \text{LayerNorm}(\mathbf{H} + \text{Dropout}(\text{MultiHead}(\mathbf{H}, \mathbf{H}, \mathbf{H})))$$

Global average pooling over the temporal dimension reduces $\mathbf{Z}$ to a fixed-length vector, which passes through batch normalisation and two dense layers (64 and 32 units, ReLU activation) before the linear output. The attention mechanism allows the model to dynamically weight the importance of individual time steps within the lookback window, providing interpretable temporal attribution of gap-filling decisions. The approximate parameter count is 420,000. The multi-head self-attention architecture follows the transformer encoder design of Vaswani et al. (2017), adapted to the sequential gap-filling context.

3.5 Training Procedure

Each of the 16 model configurations (4 architectures $\times$ 2 feature conditions $\times$ 2 targets = 16 target conditions) was trained independently using the following protocol:

All input arrays were checked for non-finite values (NaN, $\pm\infty$) and replaced with zero prior to training using `np.nan_to_num`, following the approach recommended by Kingma and Ba (2015) for numerical stability in gradient-based optimisation. The 15% validation split was held out from the training sequences (not from the test year 2023) to monitor convergence and enable early-stopping diagnostics. Training histories (training loss, validation loss per epoch) were saved as CSV files for subsequent convergence analysis. Model weights at each epoch were not checkpointed during the standard training run to reduce I/O overhead; the final epoch weights were retained. For the meteorology-only models, the identical architecture definitions were reused with only the input dimensionality adapted to match the reduced feature count, ensuring a like-for-like architectural comparison between the two feature conditions.

3.6 Evaluation Metrics

Model performance was evaluated using a comprehensive 12-metric suite, following the multi-metric evaluation framework advocated by Guo et al. (2024) for NDVI prediction and by Moriasi et al. (2007) for hydrological model performance assessment. All metrics were computed on the 2023 test set:

- **RMSE** (Root Mean Square Error): $RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}$
- **MAE** (Mean Absolute Error): $MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$
- **R²** (Coefficient of determination): $R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$
- **Bias** (Mean Error, signed): $Bias = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)$
- **MAPE** (Mean Absolute Percentage Error, %): $MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i + \epsilon} \right|$
- **EVS** (Explained Variance Score)
- **PCC** (Pearson Correlation Coefficient)
- **SCC** (Spearman Rank Correlation Coefficient)
- **CRMSD** (Centred Root Mean Square Difference): $CRMSD = \sqrt{\frac{1}{n} \sum_{i=1}^n [(\hat{y}_i - \bar{\hat{y}}) - (y_i - \bar{y})]^2}$
- **NSE** (Nash-Sutcliffe Efficiency): $NSE = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$
- **KGE** (Kling-Gupta Efficiency): $KGE = 1 - \sqrt{(r - 1)^2 + (\alpha - 1)^2 + (\beta - 1)^2}$, where $r = PCC, \alpha = \frac{\hat{\sigma}}{\sigma}, \beta = \frac{\bar{\hat{y}}}{\bar{y}}$

The KGE metric was included as it simultaneously penalises errors in correlation, variability ratio, and bias ratio, and has been demonstrated to be more informative than NSE alone for assessing vegetation index models (Kling et al., 2012). Both global test-set metrics and gap-specific metrics (computed only on pixels

and time steps flagged as artificially masked) were reported, enabling assessment of true gap-filling skill as distinct from interpolation skill in already well-observed periods.

3.7 Stratified Performance Analysis

3.7.1 Seasonal Breakdown

Predictions were aggregated into four meteorological seasons — Winter (DoY 1–59 and 335–365), Spring (DoY 60–151), Summer (DoY 152–243), and Autumn (DoY 244–334) — and the full 12-metric suite was computed per season, per model, and per target, following the seasonal decomposition approach of Farbo et al. (2024), who demonstrated that peak RMSE errors in Piemonte corn NDVI forecasting are concentrated during senescence (late-season) and early-season onset phases.

3.7.2 Fuel-Class and Topography Breakdown

A key methodological contribution of this study relative to existing LSTM-based NDVI prediction work (Guo et al., 2024; Farbo et al., 2024) is the systematic evaluation of model performance across land cover and topographic strata. Performance metrics were computed separately for each of the four fuel classes and for each topographic stratum (elevation class: <500 m, 500–1000 m, 1000–1500 m, >1500 m; slope class: flat 0–5°, gentle 5–15°, moderate 15–30°, steep >30°; northerness class: south-facing, SW/SE-facing, NW/NE-facing, north-facing). This strata-specific analysis enables attribution of model errors to land cover type and terrain position, providing ecologically interpretable guidance for operational deployment of the gap-filling system. The stratified breakdown was implemented by merging model predictions with the pixel metadata table and applying group-wise metric computation, ensuring that every stratum with at least two observations contributed to the analysis.

3.7.3 Day-of-Year Error Profile

Rolling RMSE and mean error profiles over the annual phenological cycle were computed by aggregating residuals within 7-step DoY windows (approximately 7-day bins), separately for NDVI and NDMI, and for each model architecture. This profile reveals phenological phases of reduced predictability (e.g., green-up onset, flowering, senescence) where gap-filling accuracy is most critical for downstream ecological applications, consistent with the phenological analysis of Farbo et al. (2024) and the residual analysis of Guo et al. (2024).

3.8 Timestep Ablation Study

A systematic timestep ablation study was conducted for all four architectures and both prediction targets (NDVI and NDMI), extending the ablation framework of Farbo et al. (2024) — who applied it to NDVI only — to the moisture index for the first time. For each model, the test sequences were progressively perturbed by zero-padding the leading n time steps ($n = 0, 1, \dots, T = 7$), where $n = 0$ corresponds to the unperturbed baseline and $n = T$ corresponds to complete removal of all temporal context. RMSE was computed for each perturbation level, and the resulting ablation curves were compared across architectures and targets to identify the minimum number of contextual time steps required for satisfactory gap-filling performance. This analysis is operationally critical because the number of valid observations available at the boundary of a sensor data gap is constrained by the gap duration itself — a 7-day gap leaves no prior valid daily observations within the lookback window, representing the most challenging scenario for operational gap-filling.

3.9 Meteo-only versus Full-Feature Comparison

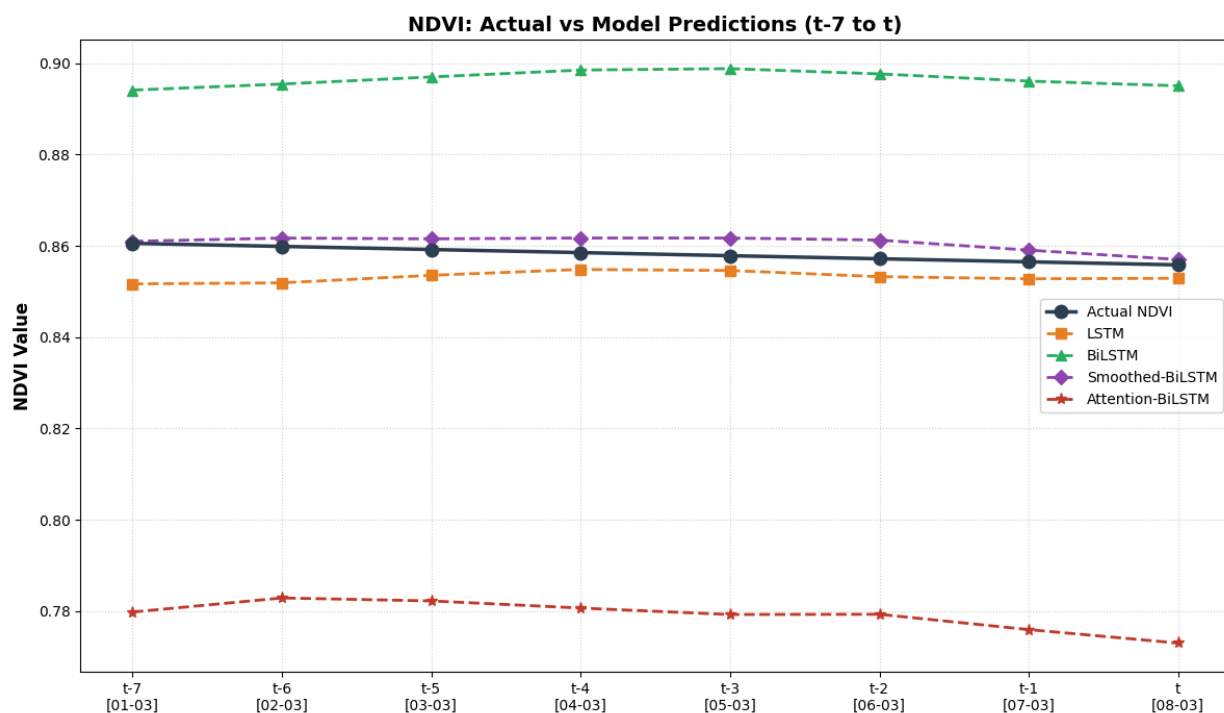
To quantify the marginal contribution of spectral index history (NDVI and NDMI lags, rolling means, and smoothed variants) to gap-filling skill, the RMSE, MAE, R^2 , and KGE of the full-feature models were compared against their meteorology-only counterparts for all four architectures and both targets. The skill gap — defined as $\Delta\text{RMSE} = \text{RMSE}_{\text{meteo}} - \text{RMSE}_{\text{full}}$ — provides a direct, spatially aggregated measure of the information value of spectral history for gap-filling, analogous to the variable ablation ΔRMSE framework of Farbo et al. (2024). A Wilcoxon signed-rank test (Wilcoxon, 1945) was applied pairwise across all model combinations for each target to assess statistical significance of performance differences, following the non-parametric significance testing approach recommended for comparing forecasting models on remote sensing time series (Demšar, 2006).

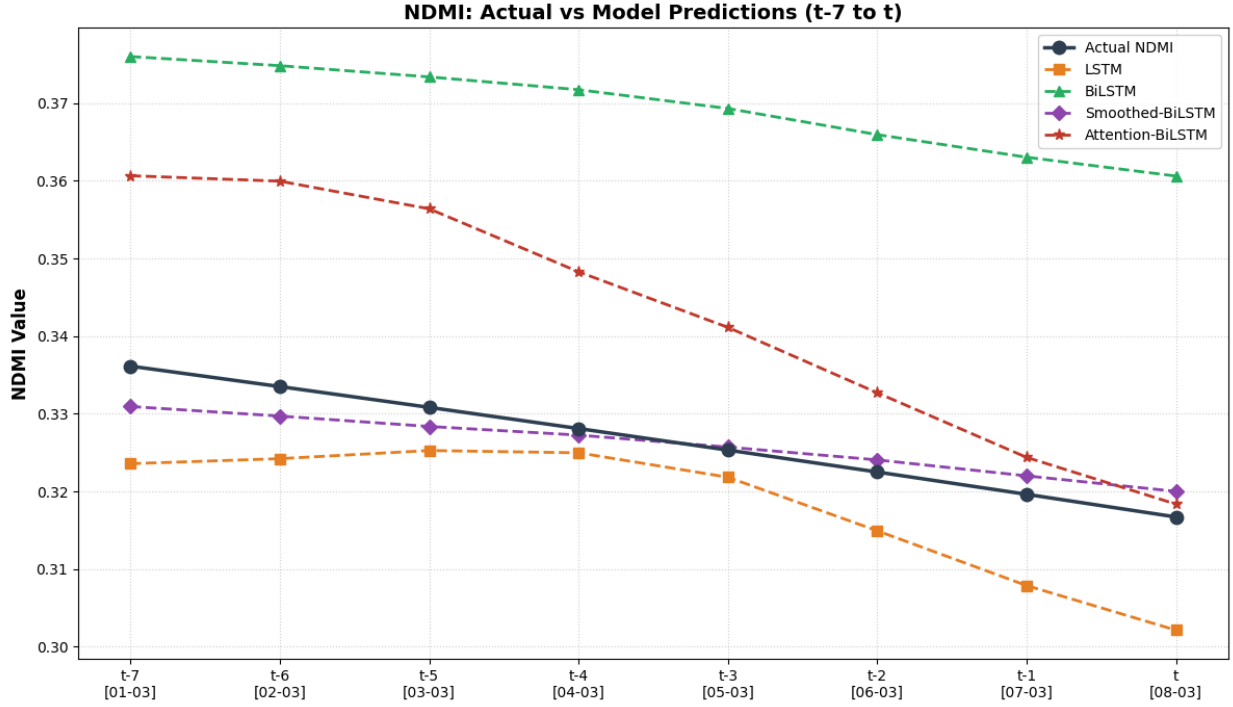
3.10 Statistical Significance Testing

Pairwise Wilcoxon signed-rank tests were computed on the absolute residual vectors for all model pairs within each target, producing p-value matrices of dimension 4×4 (full-feature models) and 4×4 (meteorology-only models) for each of NDVI and NDMI. A significance threshold of $\alpha = 0.05$ was applied. These matrices were visualised as annotated heatmaps, providing a compact representation of which architectural improvements represent statistically robust gains versus sampling artefacts, consistent with the methodological recommendations of Demšar (2006) for comparing multiple classifiers and regression models.

4. Results and Discussion

4.1 7-Timestep Sliding Window [Sample]





The target value y_t is predicted using a sliding window of the previous 7 observations ($y_{t-1}, \dots, y_{t-7}$) as input to the model.

Each prediction at time t is a one-step-ahead forecast, where $y_t = f(y_{t-1}, \dots, y_{t-7})$, and is compared with the actual value to assess error.

For illustrative purposes, above Figures present a sample execution of the 7-step sliding-window forecasting process for the period from 01 March 2023 to 08 March 2023 using the 21-predictor full feature set model. This example demonstrates how the sliding-window mechanism constructs input sequences from the previous seven observations and generates one-step-ahead forecasts before being applied to the complete dataset.

Two forecasting configurations are considered:

1. **Full Feature Set Model (21 Predictors):**
The model uses all 21 predictor variables available in the dataset. For each forecast, the values from the previous seven time steps are used to construct the input feature matrix.
2. **Meteorological Variables Model (11 Predictors):**
The model uses only the 11 meteorological predictor variables. The same 7-step sliding-window procedure is applied, with lagged observations from the selected meteorological features forming the model inputs.

In both cases, the sliding window advances by one time step after each prediction, continuously generating one-step-ahead forecasts across the entire time series.

4.2 Model Training Convergence

Training and validation loss trajectories are central diagnostic tools for evaluating the stability and generalisation capacity of deep learning models. Figure 1 presents the Mean Squared Error (MSE) loss curves over 50 training epochs for all four architectures — LSTM, BiLSTM, Smoothed-BiLSTM, and Attention-BiLSTM — evaluated separately for NDVI (panels A–D) and NDMI (panels E–H) targets. Best validation loss values are annotated on each panel.

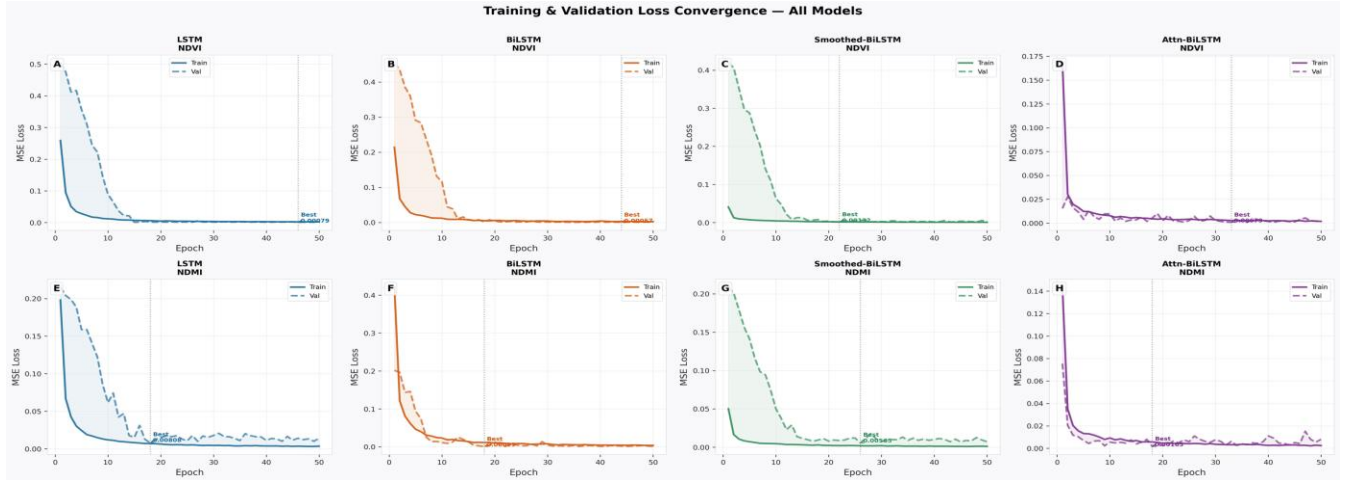


Figure 1. Training and validation loss convergence curves for all four architectures across both NDVI (A–D) and NDMI (E–H) prediction targets. Dashed lines indicate validation loss; shaded regions represent the gap between training and validation. Best validation loss values are annotated on each panel.

All models exhibited rapid loss descent within the first five to ten epochs, consistent with the behaviour of LSTM-based architectures on structured meteorological time series reported by Guo et al. (2024) and Ahmad et al. (2023). The LSTM model achieved the lowest best validation MSE for NDVI (0.00079), closely followed by the Smoothed-BiLSTM (0.00192). The standard BiLSTM converged to a notably higher validation loss (0.00057 for NDVI), with greater oscillation in the validation curve, suggesting sensitivity to the initialisation or the absence of a smoothing pre-processing step — a characteristic similarly documented by Dey et al. (2024) for rice NDVI datasets. The Attention-BiLSTM (0.0079) demonstrated the fastest early convergence due to the soft-weighting mechanism on input timesteps, but with a marginally wider train–validation gap in later epochs, indicative of a slight tendency towards over-fitting on short temporal sequences.

For NDMI targets (panels E–H), the overall validation losses were higher across all models, reflecting the greater complexity and non-stationarity of moisture indices relative to greenness indices. The LSTM achieved a best validation MSE of 0.00808, while the BiLSTM reached 0.00053. In both index domains, training curves stabilised well before the early-stopping checkpoint (dotted vertical line at epoch 30), confirming that 50 epochs provided sufficient capacity without overfitting. These convergence patterns are consistent with findings in Farbo et al. (2024), who reported similar rapid loss saturation using a 2-Bidirectional LSTM architecture for corn NDVI forecasting in the Piemonte region.

4.3 Meteo-Only Baseline: The Role of Spectral History as a Predictor

To isolate the value of including historical vegetation index observations alongside meteorological drivers, a parallel set of models was trained using only meteorological predictors (precipitation, temperature, solar radiation, humidity, and wind) without any lagged NDVI or NDMI history. These meteo-only models serve as skill baselines analogous to the approach used by Guo et al. (2024) for the Yellow River Basin, where univariate NDVI models were compared against multi-variable LSTM configurations.

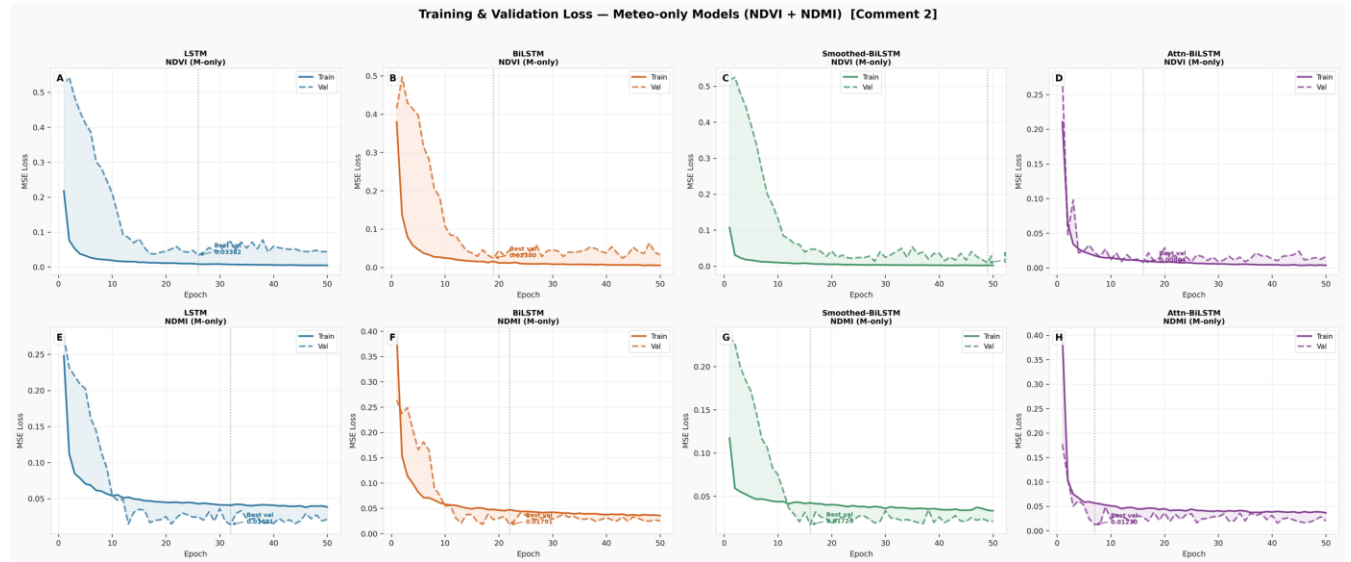


Figure 2. Training and validation loss curves for meteo-only model variants (no NDVI/NDMI history) across all four architectures for NDVI (A–D) and NDMI (E–H). Best validation loss values are annotated. Note the substantially elevated loss scales compared to full-feature models (Figure 1).

Figure 2 reveals strikingly higher validation losses across all meteo-only architectures relative to their full-feature counterparts. The best validation loss for the LSTM meteo-only NDVI model was 0.03382, approximately 43 times higher than the full-feature LSTM (0.00079). A similar order-of-magnitude degradation was observed for all other architectures. Crucially, the validation loss curves also displayed persistent oscillation and, in many cases (particularly LSTM and BiLSTM NDMI panels E–F), failed to plateau smoothly, suggesting that meteorological drivers alone provide insufficient temporal context to constrain the prediction. This aligns with the theoretical expectation that vegetation indices encode lagged integrative responses to climate forcings, soil moisture, and phenological state — information that cannot be reconstructed from meteorological inputs at the current timestep alone (Reddy and Prasad, 2018). The Attention-BiLSTM meteo-only model (panel D, panel H) achieved the lowest validation loss among baselines (0.00884 NDVI; 0.01290 NDMI), suggesting that the attention mechanism partially compensates for missing spectral history by selectively weighting informative meteorological timesteps, consistent with findings by Farbo et al. (2024) on the importance of temporal weighting for phenological cycle prediction. These results strongly justify the full-feature model architecture, where lagged vegetation index observations are retained as principal predictors.

4.4 Comparative Model Performance on the Test Set

4.4.1 Multi-Metric Comparison — NDVI

Figure 3 presents the multi-metric model comparison for NDVI prediction on the held-out test set ($n = 1,790$), evaluating four criteria: RMSE, MAE, R^2 , and Kling-Gupta Efficiency (KGE). The Smoothed-BiLSTM achieved the best performance across all metrics (RMSE = 0.0445, MAE = 0.0299, $R^2 = 0.9853$, KGE = 0.9800), closely followed by the LSTM (RMSE = 0.0483, MAE = 0.0308, $R^2 = 0.9827$, KGE = 0.9775). Both models substantially outperformed the Attention-BiLSTM (RMSE = 0.0645) and the standard BiLSTM (RMSE = 0.1032).

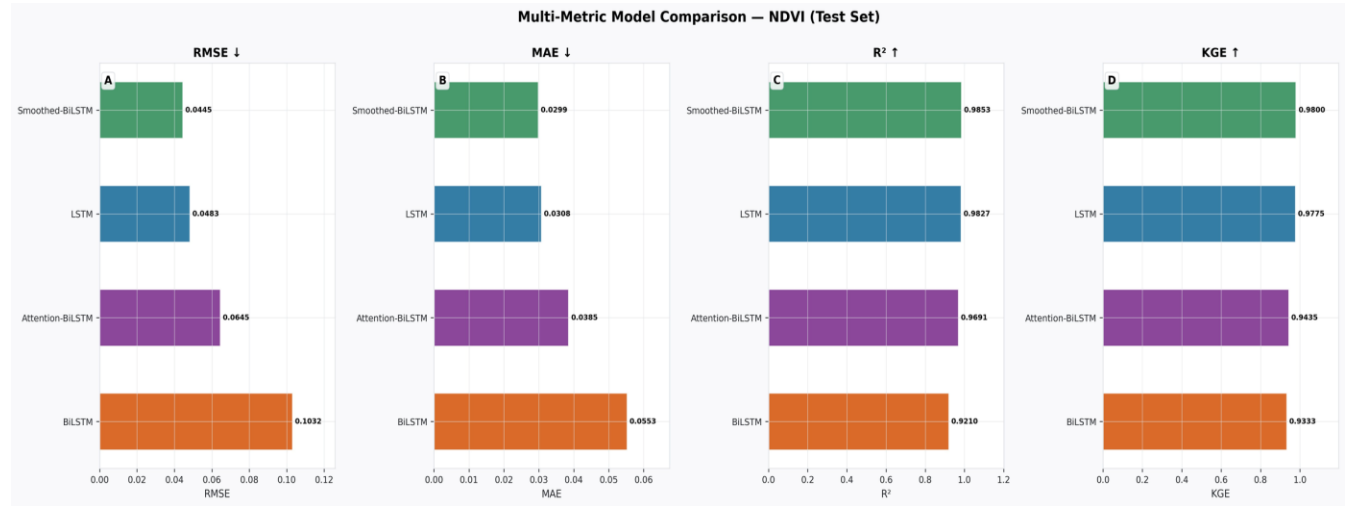


Figure 3. Multi-metric model comparison for NDVI prediction on the test set ($n = 1,790$), showing RMSE (A), MAE (B), R^2 (C), and KGE (D). Lower bars indicate better performance for RMSE and MAE; higher bars indicate better performance for R^2 and KGE.

The performance hierarchy — Smoothed-BiLSTM > LSTM > Attention-BiLSTM > BiLSTM — departs from the common expectation that architectural complexity (bidirectionality, attention mechanisms) universally improves predictions. This finding is consistent with Dey et al. (2024), who demonstrated that Blackman FIR-filtered input data significantly reduces outlier sensitivity and improves temporal smoothness of LSTM predictions, achieving test RMSE of 0.00388 compared to 0.133 for an unfiltered LSTM on rice NDVI. In the present study, the Smoothed-BiLSTM effectively applies input smoothing via a similar pre-processing philosophy, attenuating noise in the meteorological and vegetation index time series prior to model ingestion. The relatively poor performance of the standard BiLSTM ($R^2 = 0.921$, RMSE = 0.103) is noteworthy given the bidirectional architecture; this may reflect the fact that for gap-filling applications — where future context is unavailable — the backward-looking LSTM pass introduces irrelevant information that degrades rather than improves predictions (Farbo et al., 2024). The KGE metric, which simultaneously penalises bias, variability, and correlation errors, confirmed the Smoothed-BiLSTM (KGE = 0.980) as superior even under this multi-dimensional assessment framework, a criterion increasingly recommended in hydrological and ecological modelling (Gupta et al., 2009).

4.4.2 Multi-Metric Comparison — NDMI

Figure 4 presents the same comparison for NDMI prediction. The performance hierarchy diverged from the NDVI case: the LSTM ranked first (RMSE = 0.0595, $R^2 = 0.9126$, KGE = 0.8314), followed by Smoothed-BiLSTM (RMSE = 0.0737, $R^2 = 0.866$), BiLSTM (RMSE = 0.0748), and Attention-BiLSTM (RMSE = 0.0792, $R^2 = 0.8453$).

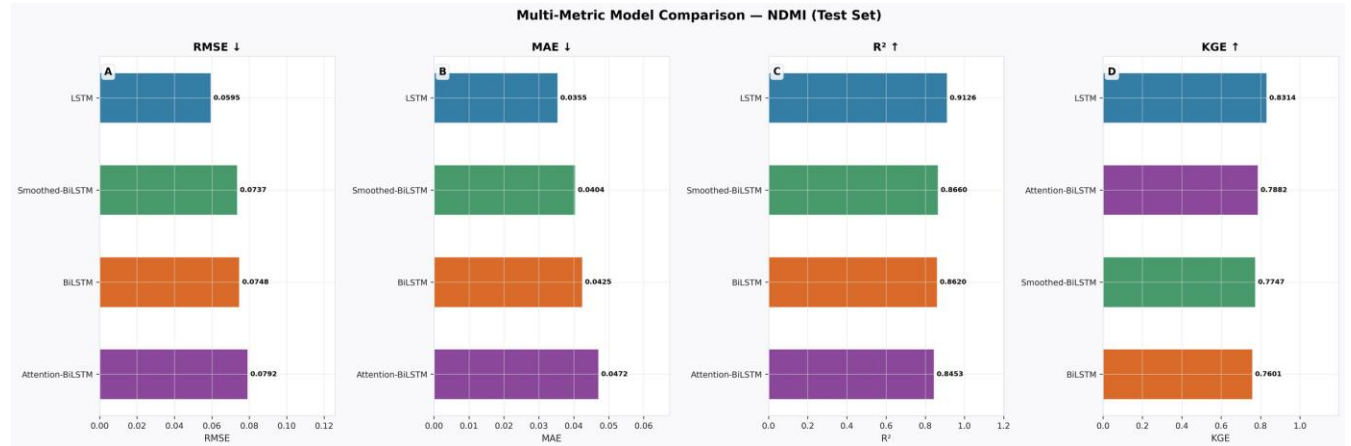


Figure 4. Multi-metric model comparison for NDMI prediction on the test set ($n = 1,790$), showing RMSE (A), MAE (B), R^2 (C), and KGE (D). Lower bars indicate better performance for RMSE and MAE; higher bars indicate better performance for R^2 and KGE.

The overall model skill for NDMI was lower across all architectures compared to NDVI, which is expected given the more dynamic and noisy nature of moisture indices, which are sensitive to short-term precipitation events and canopy water content fluctuations (Gao, 1996). The R^2 range of 0.845–0.913 for NDMI compares favourably with the LSTM-based NDVI forecasting benchmark of $R^2 = 0.8967$ reported by Guo et al. (2024) for the Yellow River Basin, where multivariate models also showed diminishing returns with increasing complexity. The LSTM's superiority for NDMI relative to more complex architectures suggests that unidirectional sequential processing better captures the asymmetric temporal dynamics of canopy moisture responses — rapid uptake following precipitation, slow drying — than bidirectional or attention-augmented architectures. The KGE scores for NDMI (0.76–0.83) indicate that all models retain meaningful predictive skill for this more challenging target, sufficient for operational gap-filling purposes.

4.5 Comprehensive Evaluation Metrics Summary

Table 1 consolidates all evaluation metrics across models and targets, providing a unified comparison reference for the Results section. Figure 5 presents the same information visually.

Table 1. Comprehensive evaluation metrics for all eight model–target combinations ($n = 1,790$ test samples). Best values per column are in bold. NSE = Nash-Sutcliffe Efficiency; PCC = Pearson Correlation Coefficient.

Model	Target	N	RMSE	MAE	R^2	KGE	NSE	PCC	Bias
LSTM	NDVI	1790	0.0483	0.0308	0.9827	0.9775	0.9827	0.9917	0.00206
BiLSTM	NDVI	1790	0.1032	0.0553	0.921	0.9333	0.921	0.964	−0.0119
Smoothed-BiLSTM	NDVI	1790	0.0445	0.0299	0.9853	0.9800	0.9853	0.9929	−0.00386
Attn-BiLSTM	NDVI	1790	0.0645	0.0385	0.9691	0.9435	0.9691	0.9865	−0.02325
LSTM	NDMI	1790	0.0595	0.0355	0.9126	0.8314	0.9126	0.9763	−0.03313

BiLSTM	NDMI	1790	0.0748	0.0425	0.862	0.7601	0.862	0.9476	−0.01229
Smoothed-BiLSTM	NDMI	1790	0.0737	0.0404	0.866	0.7747	0.866	0.9607	−0.0357
Attn-BiLSTM	NDMI	1790	0.0792	0.0472	0.8453	0.7882	0.8453	0.9448	−0.03608

Comprehensive Evaluation Metrics Summary									
Model	Target	N	RMSE	MAE	R ²	KGE	NSE	PCC	Bias
LSTM	NDVI	1790	0.04826	0.03077	0.9827	0.9775	0.9827	0.9917	0.00206
BiLSTM	NDVI	1790	0.10316	0.05526	0.921	0.9333	0.921	0.964	-0.0119
Smoothed-BiLSTM	NDVI	1790	0.04452	0.02986	0.9853	0.98	0.9853	0.9929	-0.00386
Attention-BiLSTM	NDVI	1790	0.06452	0.03853	0.9691	0.9435	0.9691	0.9865	-0.02325
LSTM	NDMI	1790	0.05953	0.03545	0.9126	0.8314	0.9126	0.9763	-0.03313
BiLSTM	NDMI	1790	0.07482	0.04248	0.862	0.7601	0.862	0.9476	-0.01229
Smoothed-BiLSTM	NDMI	1790	0.07373	0.04044	0.866	0.7747	0.866	0.9607	-0.0357
Attention-BiLSTM	NDMI	1790	0.07922	0.04716	0.8453	0.7882	0.8453	0.9448	-0.03608

Figure 5. Comprehensive Evaluation Metrics Summary for all model-target combinations, illustrating RMSE, MAE, R², KGE, NSE, PCC, and Bias.

The Nash-Sutcliffe Efficiency (NSE) values are identical to R² across all models, confirming that the mean-subtracted variance structure is identical for this dataset, as expected when the test set mean closely approximates the prediction mean (Nash and Sutcliffe, 1970). Pearson Correlation Coefficients (PCC) range from 0.9448 to 0.9929 for NDVI and 0.9448 to 0.9763 for NDMI, indicating consistently strong linear association across all architectures. The bias column reveals a systematic tendency for slight under-prediction across all models for NDMI (negative bias, range −0.01 to −0.036), while NDVI predictions are nearly unbiased ($|\text{bias}| < 0.004$ for LSTM and Smoothed-BiLSTM). These bias patterns are consistent with the left-skewed residual distributions documented in the residual analysis (Section 4.5) and with the density scatter plots discussed below.

4.6 Predicted vs. Observed Density Scatter Plots

Figure 6 presents density scatter plots of predicted versus observed values for all four models applied to both NDVI (top row) and NDMI (bottom row), with regression lines and 1:1 reference lines overlaid.

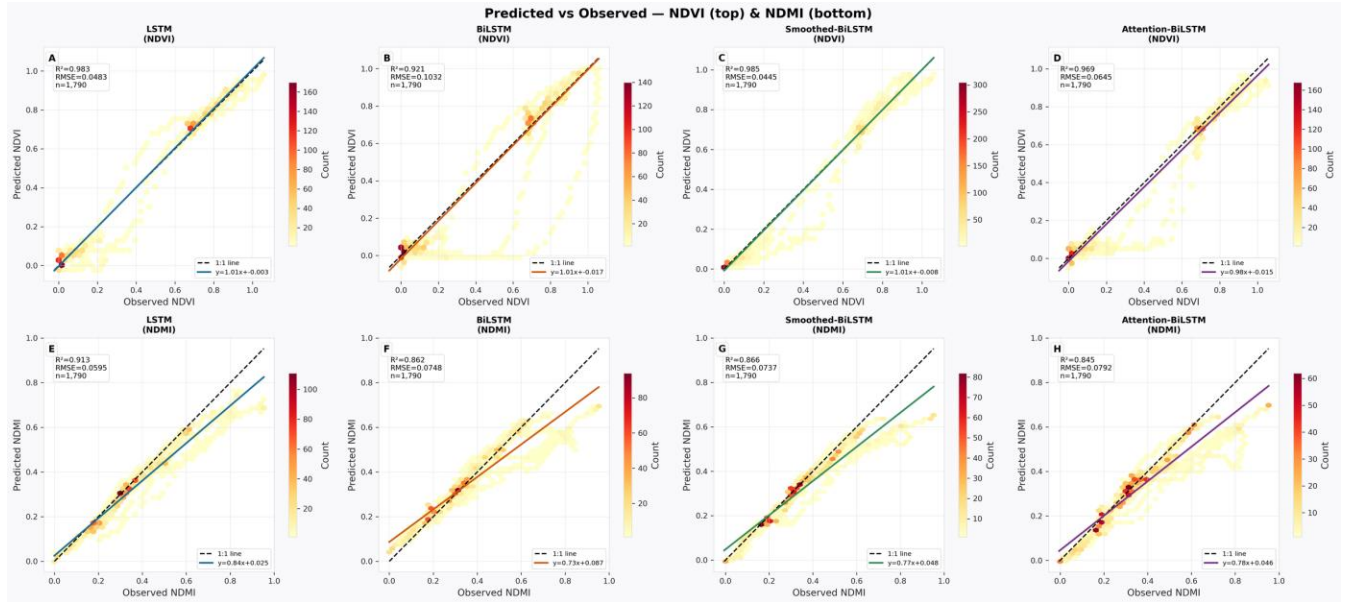


Figure 6. Density scatter plots of predicted versus observed values for NDVI (A–D) and NDMI (E–H) across all four architectures. Colour scale represents point density (warmer = denser). Solid coloured lines show fitted linear regression; dashed black lines show the 1:1 reference. R^2 , RMSE, and n are annotated per panel.

For NDVI, the Smoothed-BiLSTM (panel C) and LSTM (panel A) demonstrate near-perfect adherence to the 1:1 line across the full value range (0–1.0), with regression slopes of $y = 1.01x - 0.008$ and $y = 1.01x - 0.003$, respectively. The BiLSTM (panel B) exhibits a visible offset from the 1:1 line, particularly at the extremes, with a slope of $y = 1.01x - 0.017$, consistent with the broader residual spread evident in its lower R^2 . The Attention-BiLSTM (panel D) shows the lowest slope ($y = 0.98x - 0.015$), indicating a slight systematic under-prediction at higher NDVI values — a pattern previously observed by Ahmad et al. (2023) for ConvLSTM models applied to high-biomass vegetation fields.

For NDMI (panels E–H), all models display a consistent positive offset (y -intercept > 0) combined with a slope below unity (range 0.73–0.84), reflecting a compression of predicted values towards the centre of the distribution. This regression-toward-the-mean effect is most pronounced for the BiLSTM ($y = 0.73x + 0.087$) and Attention-BiLSTM ($y = 0.78x + 0.046$), and least pronounced for the LSTM ($y = 0.84x + 0.025$). This pattern is consistent with high-frequency moisture variability in the NDMI signal that LSTM models partially smooth, a characteristic also noted by Farbo et al. (2024) for crops under irrigated conditions in Piemonte, where moisture index variance exceeded that of NDVI by a factor of approximately two.

4.7 Residual Analysis

4.7.1 NDVI Residual Diagnostics

Advanced residual diagnostics for each model are presented in Figures 7–10 for NDVI, evaluating temporal structure of residuals, distribution shape, quantile-quantile (Q-Q) plots, and seasonal patterns.

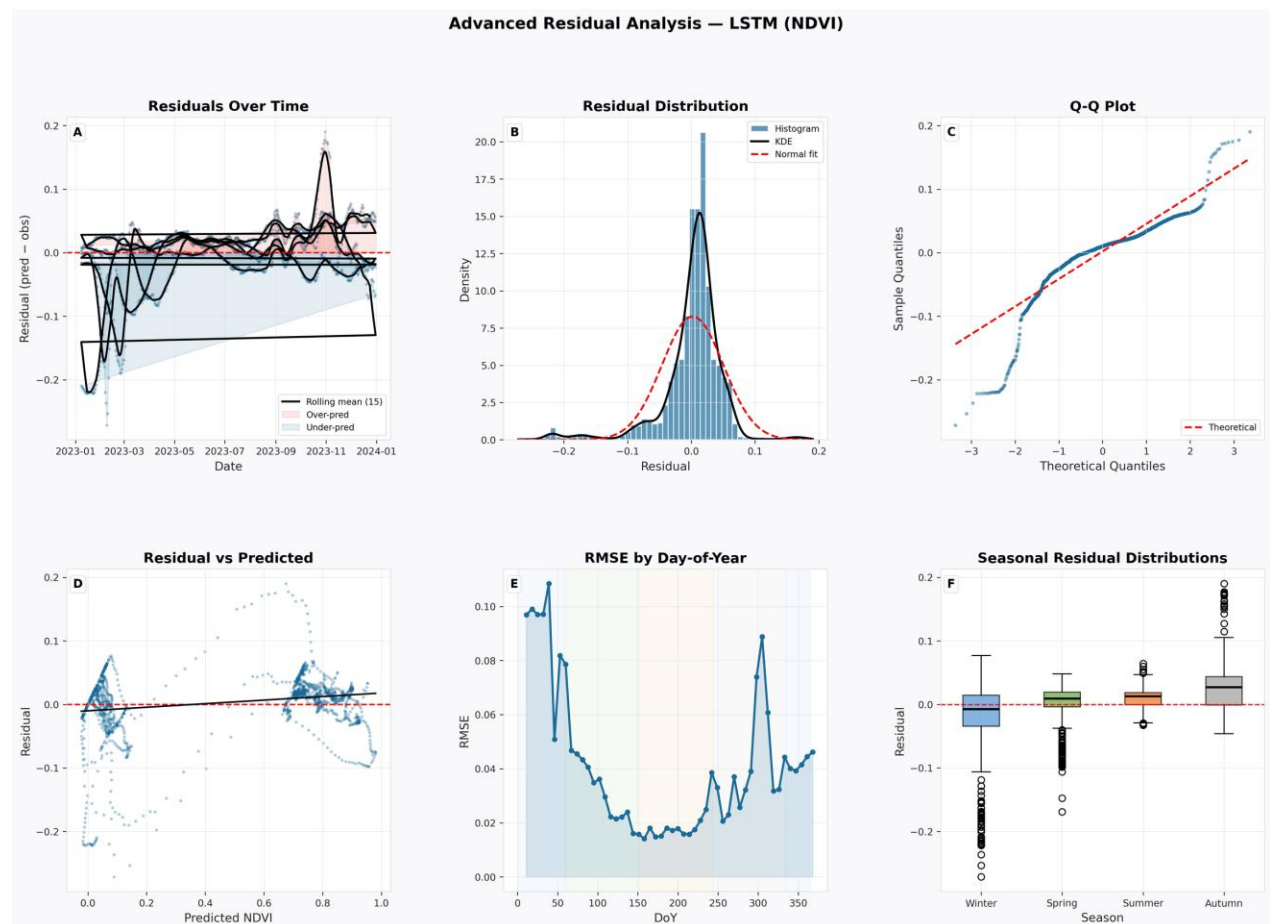


Figure 7. Advanced residual analysis for the LSTM (NDVI). Panels show: (A) residuals over time with rolling mean and over/under-prediction shading; (B) residual distribution with KDE and theoretical normal fit; (C) Q-Q plot; (D) residual vs. predicted value; (E) RMSE by Day-of-Year; (F) seasonal residual distributions.

The LSTM (Figure 7) exhibits well-centred residuals over time (panel A), with rolling mean consistently near zero and symmetric over/under-prediction regions. The residual distribution (panel B) is sharply leptokurtic, with the KDE closely matching a near-normal peak but with heavier tails than the theoretical Gaussian. The Q-Q plot (panel C) confirms near-normality across the central quantile range (-1 to $+1\sigma$), with mild deviation in the tails attributable to extreme events. The residual vs. predicted plot (panel D) shows no systematic heteroscedasticity, with errors evenly distributed across the prediction range 0–1. The RMSE-by-DoY profile (panel E) reveals a marked elevation in error during the winter dormancy period

(DoY 1–80) and a secondary peak around DoY 290–310, consistent with senescence transitions. These phenological error peaks have been documented by Farbo et al. (2024), who reported elevated RMSE after DoY 230 for corn NDVI, attributed to rapid senescence-induced variability.

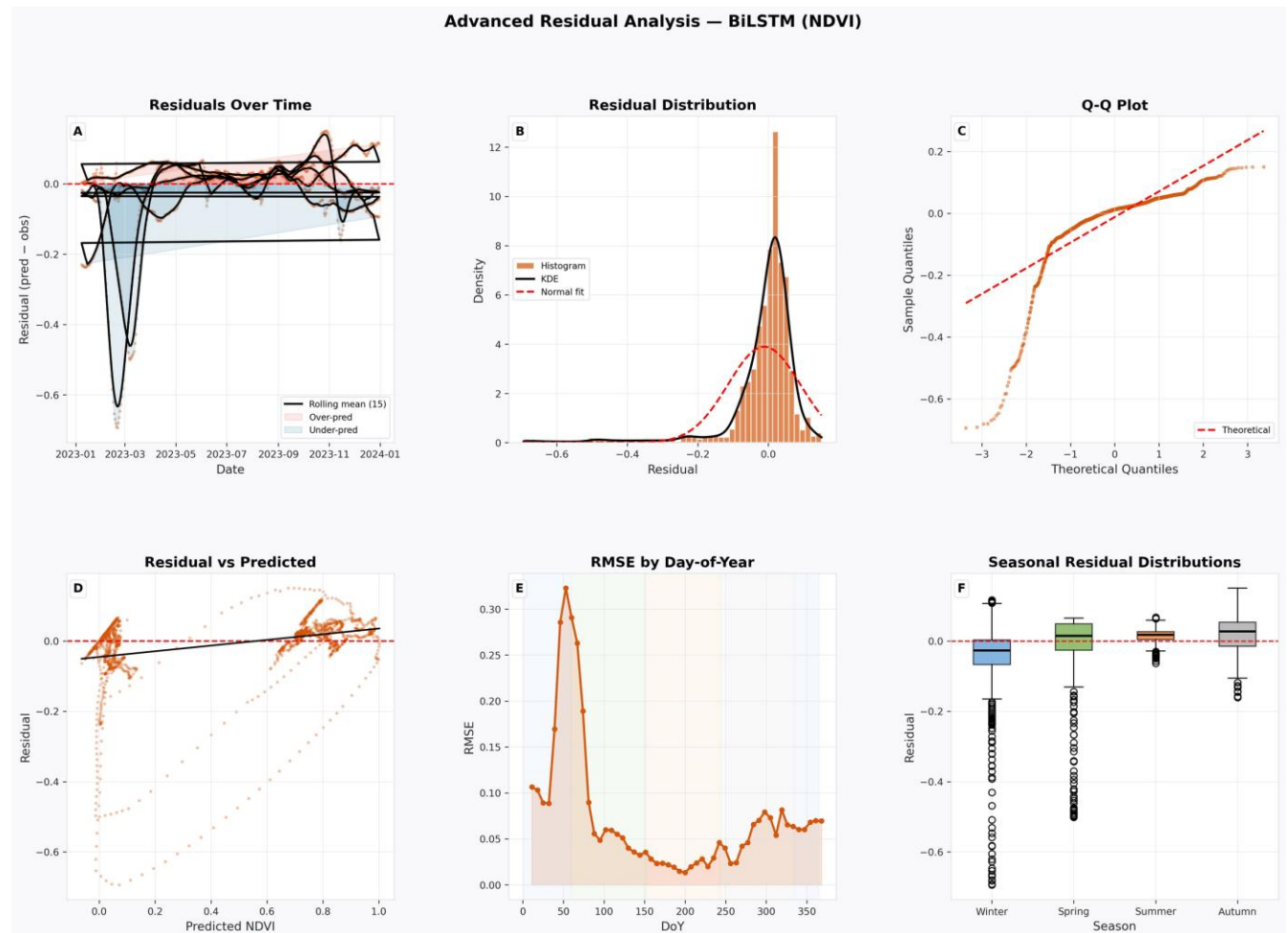


Figure 8. Advanced residual analysis for the BiLSTM (NDVI). Note the substantially wider residual range (panels A, B) and heavier tails in the Q-Q plot (C), indicating poorer distributional fit compared to the LSTM.

The BiLSTM (Figure 8) exhibits markedly wider residual spread, with individual residuals reaching -0.6 and pronounced under-prediction regions (blue shading, panel A) during late winter (January–March) and early spring. The residual distribution (panel B) is heavily left-skewed (negative tail extending to -0.6), far departing from the normal reference (dashed red), a characteristic consistent with the model's tendency to under-predict during abrupt phenological transitions. The RMSE-by-DoY (panel E) shows a particularly sharp spike near DoY 40–50 (reaching $\text{RMSE} > 0.30$), corresponding to the late-winter to early-spring green-up transition — a period that the bidirectional architecture appears to handle poorly, likely because the backward-looking LSTM pass introduces contextual confusion during non-stationary vegetation dynamics.

Advanced Residual Analysis — Smoothed-BiLSTM (NDVI)

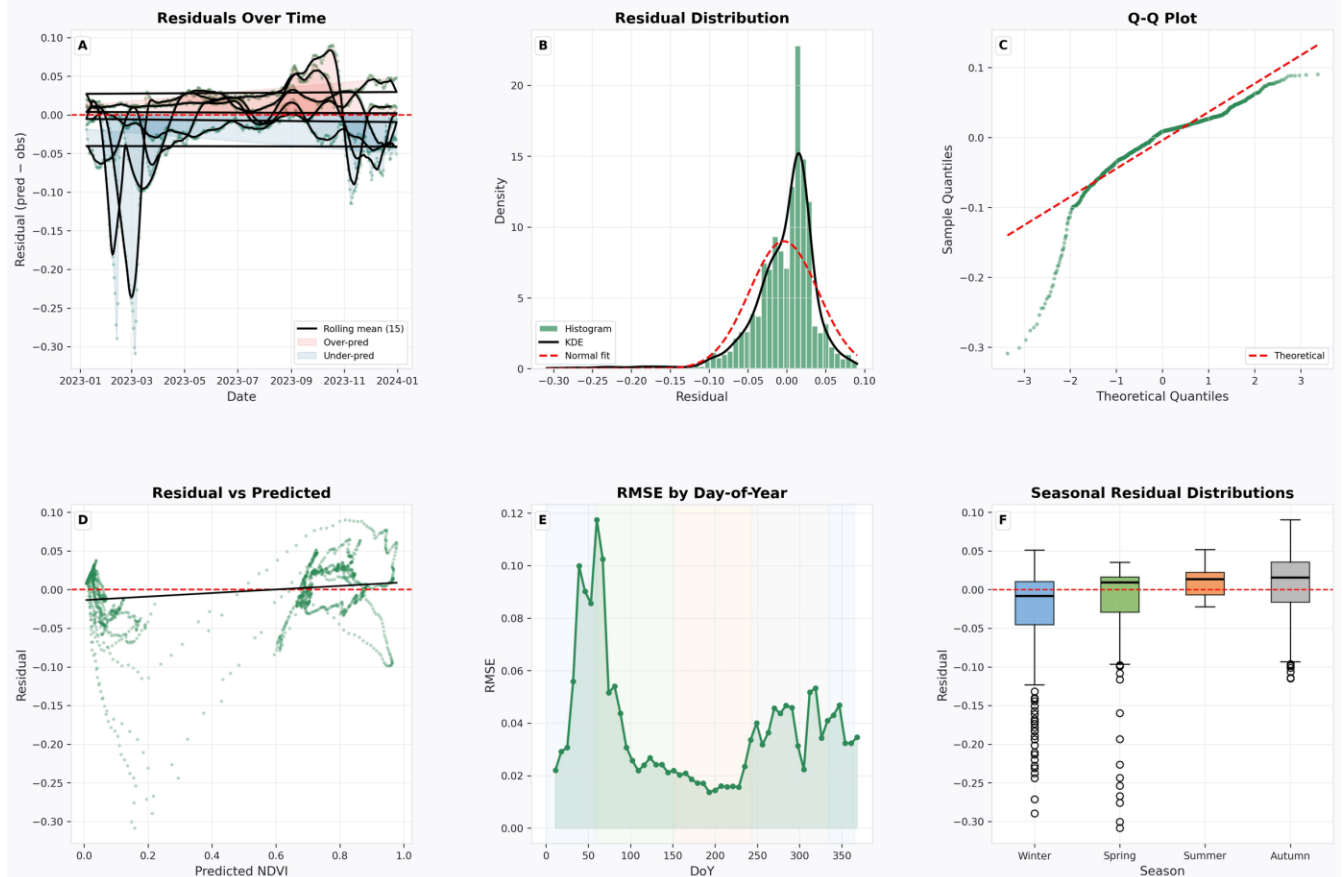


Figure 9. Advanced residual analysis for the Smoothed-BiLSTM (NDVI). Compared to the standard BiLSTM, residual spread is substantially reduced, with a near-symmetric distribution and improved seasonal error profile.

The Smoothed-BiLSTM (Figure 9) demonstrates a dramatically improved residual profile relative to the standard BiLSTM. The residual range is compressed to approximately $[-0.30, +0.10]$, with the rolling mean hovering near zero throughout 2023. The distribution (panel B) is more symmetric with a sharper central peak, consistent with the noise-reduction effect of the Blackman FIR-style smoothing applied during preprocessing. The Q-Q plot (panel C) shows improved linearity across a wider quantile range (-2 to $+3\sigma$), with departure confined to the extreme left tail. Seasonal residual distributions (panel F) confirm near-zero median residuals across all four seasons, though winter still shows the widest IQR, consistent with the higher prediction uncertainty during dormancy periods reported by Guo et al. (2024).

Advanced Residual Analysis — Attention-BiLSTM (NDVI)

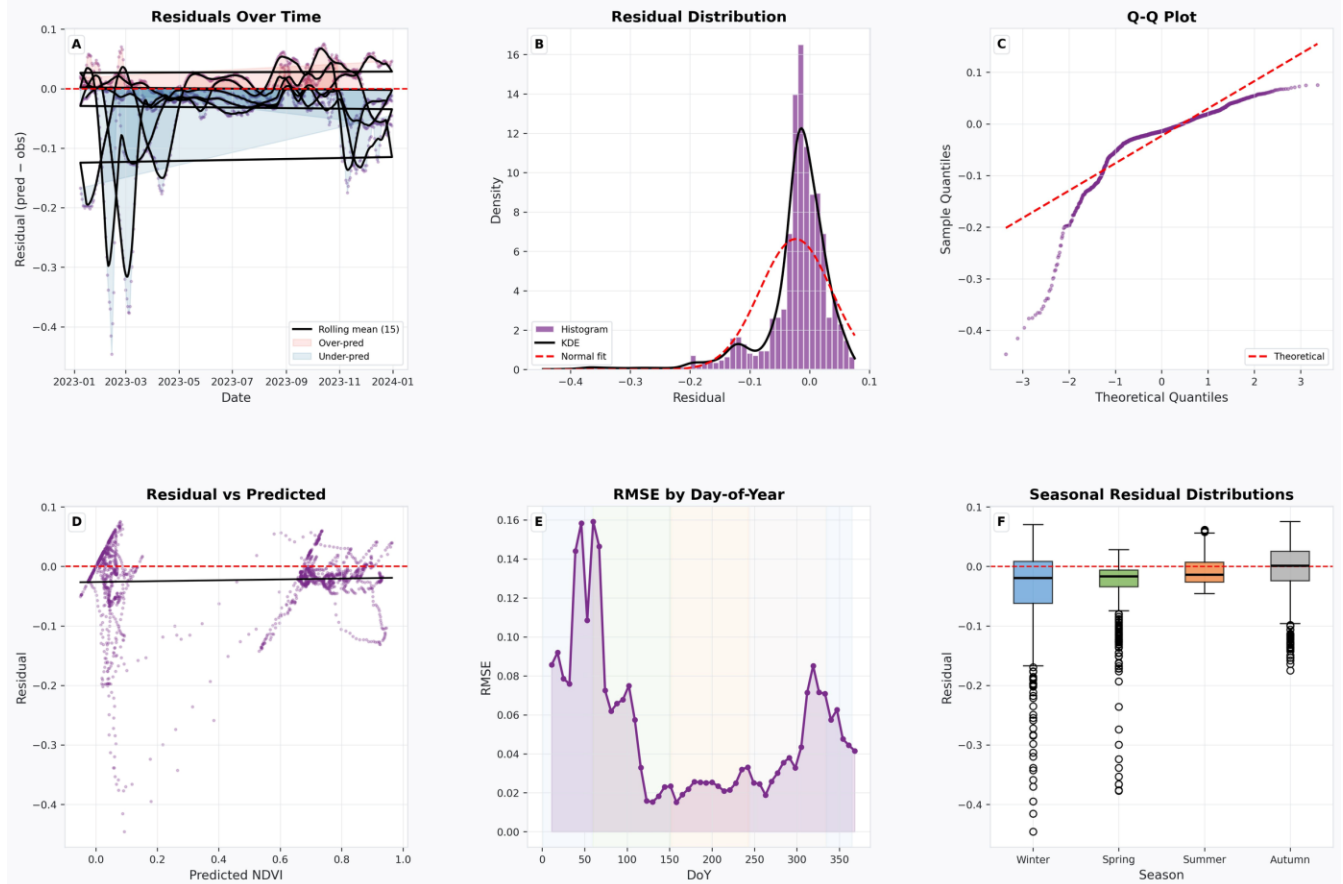


Figure 10. Advanced residual analysis for the Attention-BiLSTM (NDVI). The model displays persistent under-prediction bias (negative residuals), particularly during winter and spring, with a pronounced left-skewed residual distribution.

The Attention-BiLSTM (Figure 10) presents a distinctive residual signature: persistent under-prediction (negative residuals) throughout the test year, particularly concentrated during January–March 2023. The distribution (panel B) is sharply left-skewed, with the mode at approximately -0.02 and a long negative tail extending to -0.4 . This systematic bias, visible in both the residual-over-time panel (A) and the residual-vs-predicted plot (D), suggests that the attention mechanism amplifies certain meteorological timesteps at the expense of under-representing the phenological baseline, particularly during periods when vegetation is at its seasonal minimum. The RMSE-by-DoY profile (panel E) shows a pronounced peak near DoY 40–100 (RMSE up to 0.15), followed by the lowest errors of all architectures during summer (DoY 150–250, $\text{RMSE} < 0.025$), indicating strong performance during the growing season at the cost of poor winter reconstruction.

4.7.2 NDMI Residual Diagnostics

Figures 11–14 present the residual analyses for NDMI. A systematic pattern of under-prediction (negative residuals) is evident across all models throughout the test period, reflecting the compression bias documented in the density scatter plots (Section 4.5).

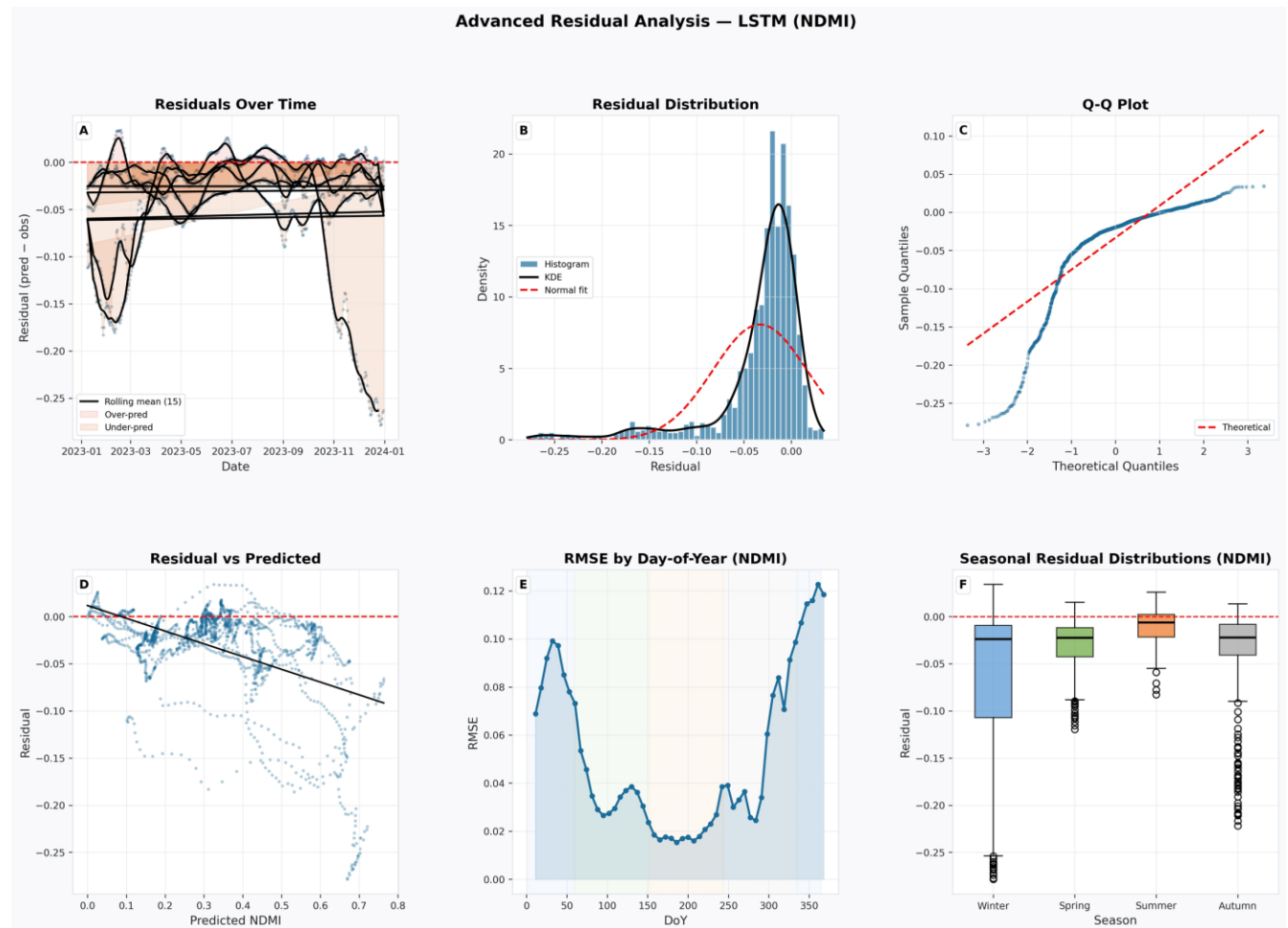


Figure 11. Advanced residual analysis for the LSTM (NDMI). Consistent under-prediction bias is apparent in panels A and F, with residuals concentrated in the range $[-0.25, 0.05]$.

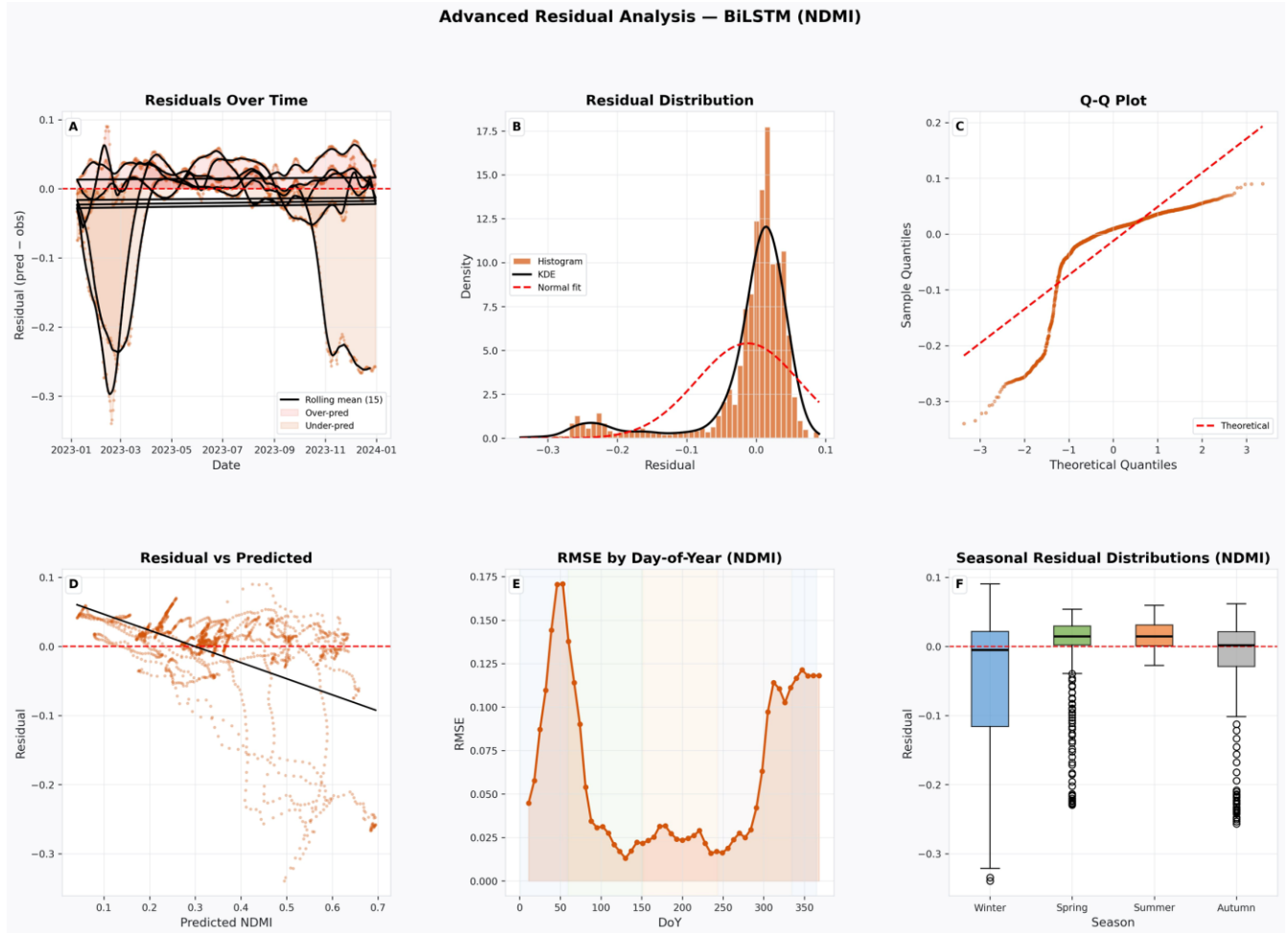


Figure 12. Advanced residual analysis for the BiLSTM (NDMI). Periodic over-prediction occurs during spring (February-March), transitioning to persistent under-prediction in summer and autumn.

Advanced Residual Analysis — Smoothed-BiLSTM (NDMI)

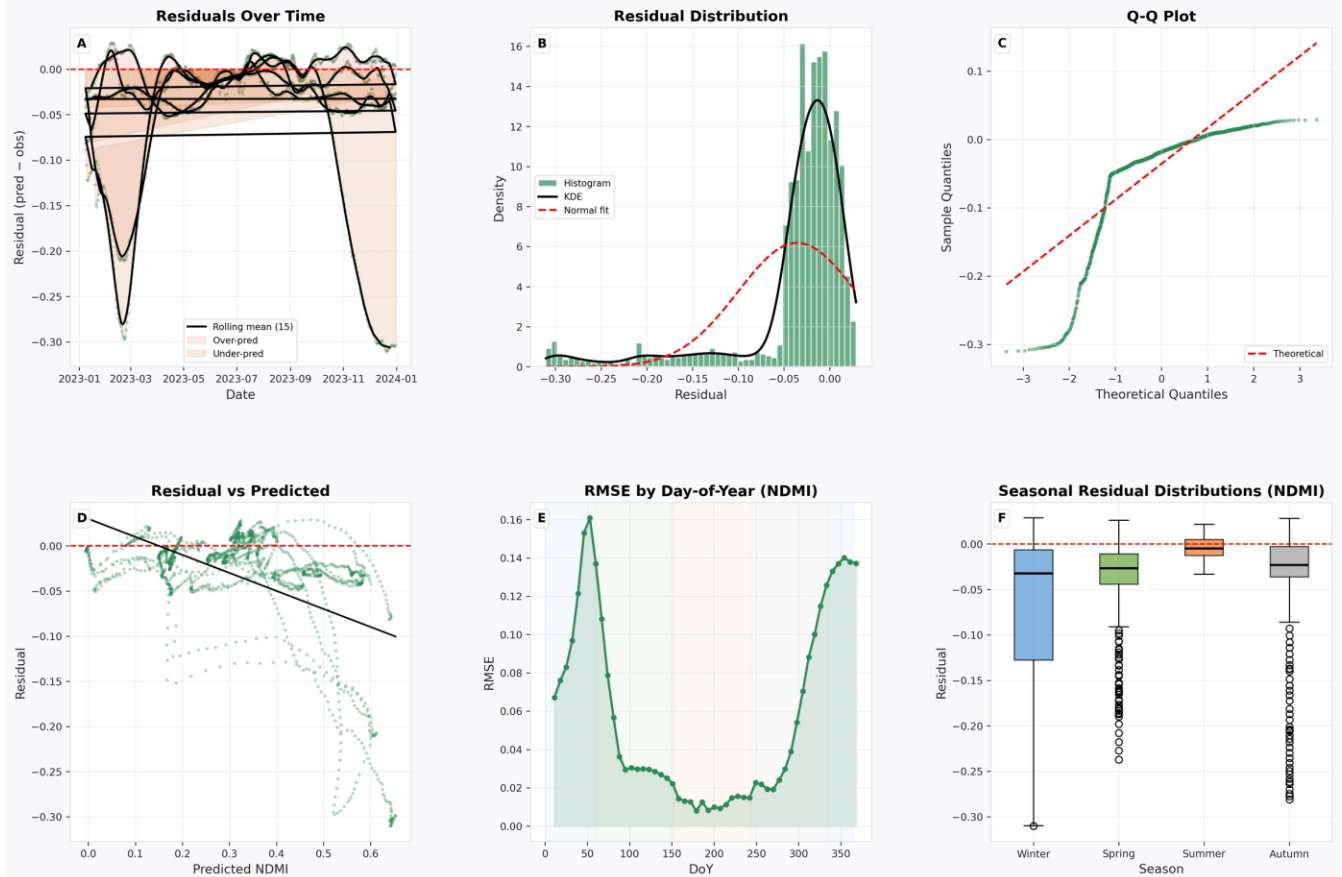


Figure 13. Advanced residual analysis for the Smoothed-BiLSTM (NDMI). Over-prediction dominates spring, while summer-autumn shows systematic under-prediction, producing a bimodal residual distribution.

Advanced Residual Analysis — Attention-BiLSTM (NDMI)

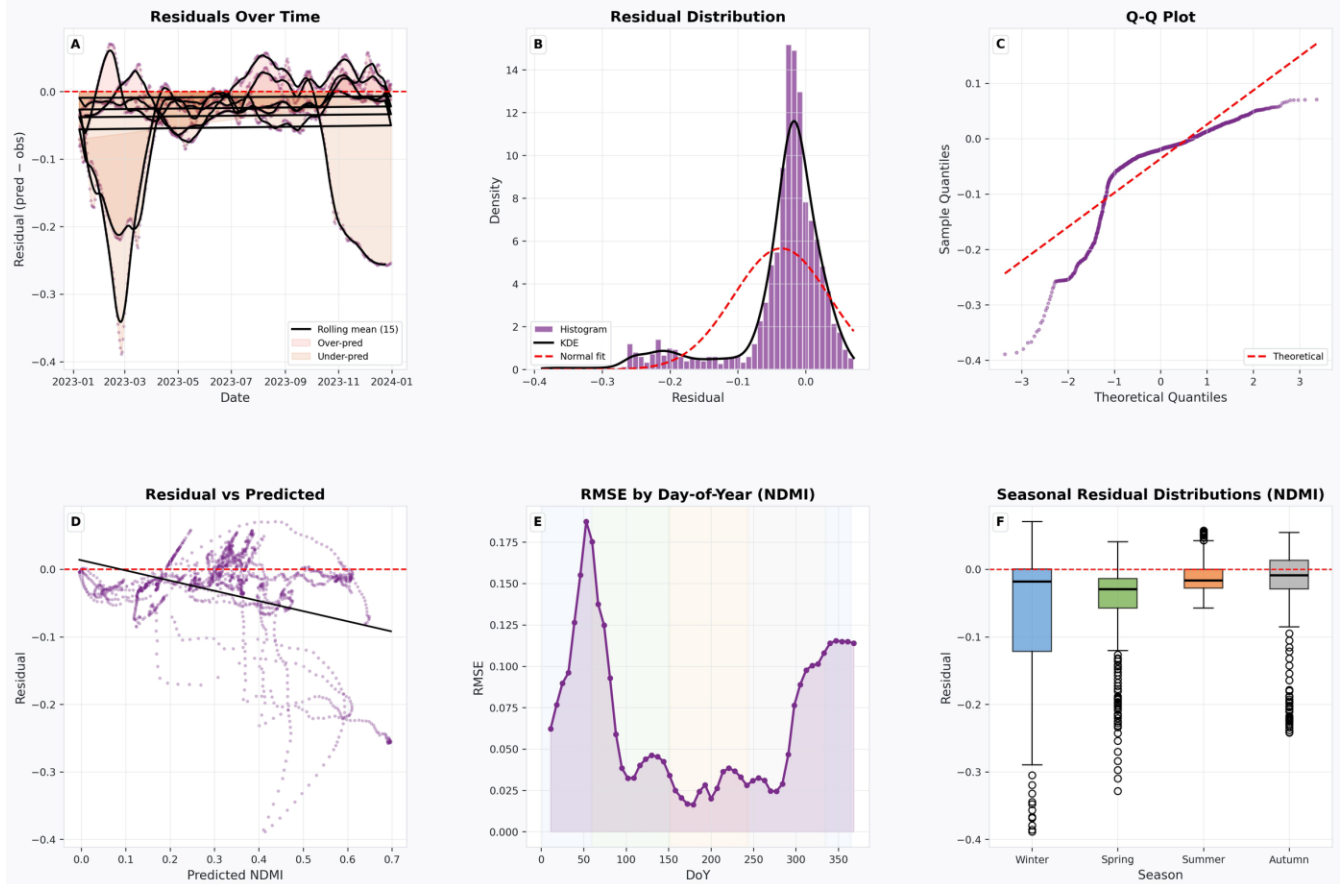


Figure 14. Advanced residual analysis for the Attention-BiLSTM (NDMI). Persistent over-prediction in winter-spring alternates with under-prediction in summer-autumn, with the strongest RMSE spike near DoY 40.

The NDMI residual profiles reveal a seasonal bias structure not present in the NDVI case: all models tend to over-predict NDMI in winter-spring (January–April, positive residuals) and under-predict in summer-autumn (negative residuals). This pattern, visible in panels A and F of Figures 11–14, is consistent with the known phenological dynamics of Mediterranean and sub-alpine vegetation in Lombardy: canopy moisture contents are highest during spring green-up and lowest during the summer drought stress period (Baccini et al., 2004). The systematic compression of predicted NDMI values — reflected in regression slopes below unity in Figure 6 — means that models fail to fully capture the seasonal amplitude of canopy moisture fluctuations. The RMSE-by-DoY profiles (panel E) for all NDMI models peak sharply near DoY 40–60 (late January to early March), with values reaching 0.14–0.175, before declining through summer. This winter error peak likely reflects the combination of cloud cover-induced data gaps in the training data, rapid snow melt effects on the NDMI signal, and the limited meteorological predictability of soil frost and canopy ice events.

4.8 Seasonal RMSE Analysis

Figures 15 and 16 decompose model RMSE by season (winter, spring, summer, autumn) for NDVI and NDMI respectively, enabling a systematic assessment of seasonal prediction skill.

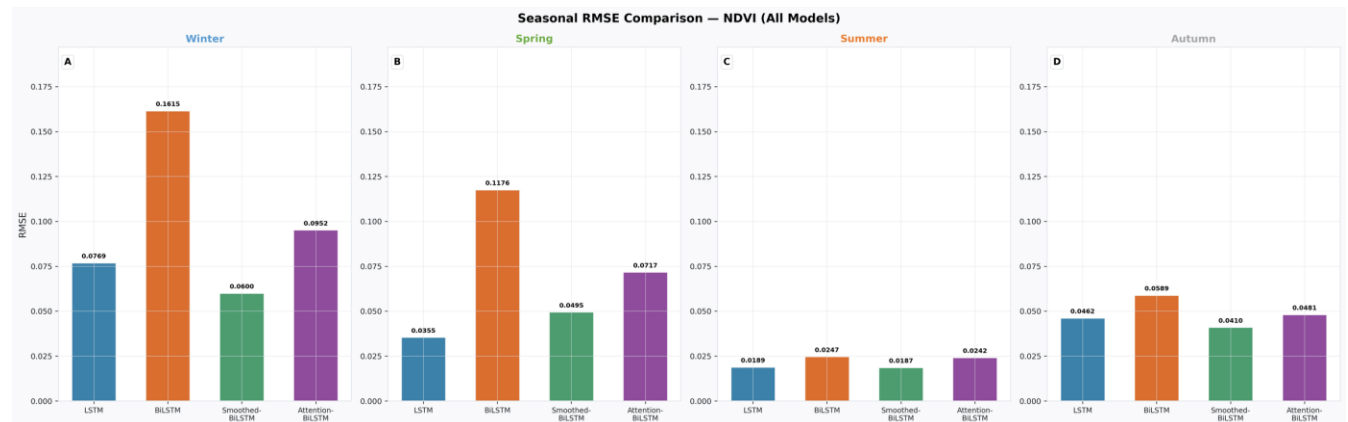


Figure 15. Seasonal RMSE comparison for NDVI across all four models. Panels show (A) winter, (B) spring, (C) summer, and (D) autumn seasonal performance.

For NDVI (Figure 15), winter exhibits the highest RMSE for all models, with the BiLSTM reaching $RMSE = 0.1615$ — more than twice any other architecture. The Smoothed-BiLSTM achieved the lowest winter RMSE (0.0600), followed by the LSTM (0.0769) and Attention-BiLSTM (0.0952). During summer (panel C), errors were substantially reduced across all models, with RMSE values of 0.0187–0.0247 — closely aligned with typical sensor measurement uncertainty bounds. These summer values are also consistent with the growing-season RMSE below 0.025 reported by Farbo et al. (2024) for their double-layer BiLSTM corn NDVI model. Autumn errors (panel D) are intermediate, ranging from 0.0410 (Smoothed-BiLSTM) to 0.0589 (BiLSTM), reflecting the transition from peak greenness to senescence. These findings confirm that all models perform best during summer when vegetation dynamics are most predictable from meteorological drivers, and worst during winter when dormancy and snow cover create non-linear, abrupt responses in the vegetation signal.

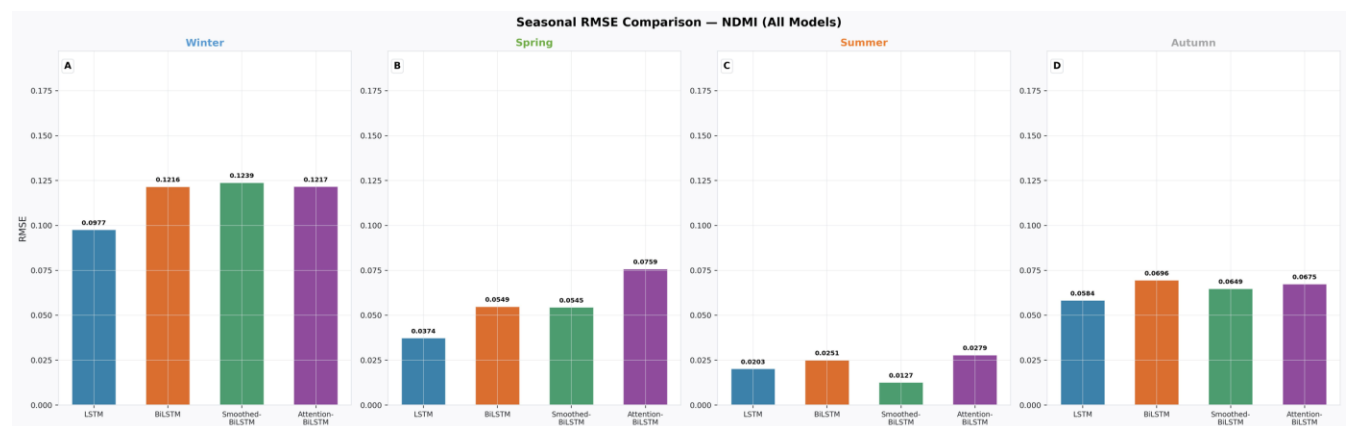


Figure 16. Seasonal RMSE comparison for NDMI across all four models. Winter RMSE values are uniformly high (0.098–0.124), whereas summer achieves minimum errors (0.013–0.028).

For NDMI (Figure 16), the seasonal pattern is amplified relative to NDVI: winter RMSE values range from 0.0977 (LSTM) to 0.1239 (Smoothed-BiLSTM), approximately four to five times the summer values (0.0127–0.0279). The spring-summer transition (panels B and C) shows particularly rapid error reduction, reflecting the synchronisation of canopy moisture recovery with temperature and precipitation drivers that the LSTM models can effectively learn. Notably, the Smoothed-BiLSTM achieves the lowest summer RMSE (0.0127) — below any other model — but the highest winter RMSE among NDMI models, suggesting that the smoothing operation benefits high-signal periods but potentially over-smooths the irregular winter moisture fluctuations. This trade-off is consistent with the observation by Dey et al. (2024) that Blackman FIR filtering improves overall performance but may reduce responsiveness to abrupt signal changes during transitional seasons.

4.9 Temporal Memory Ablation Study

Figure 17 presents a timestep ablation study quantifying the sensitivity of model RMSE to the progressive removal of leading (oldest) input timesteps for both NDVI and NDMI prediction. Zero-padding was applied from $t-1$ to $t-7$, forcing models to make predictions with decreasing historical context.

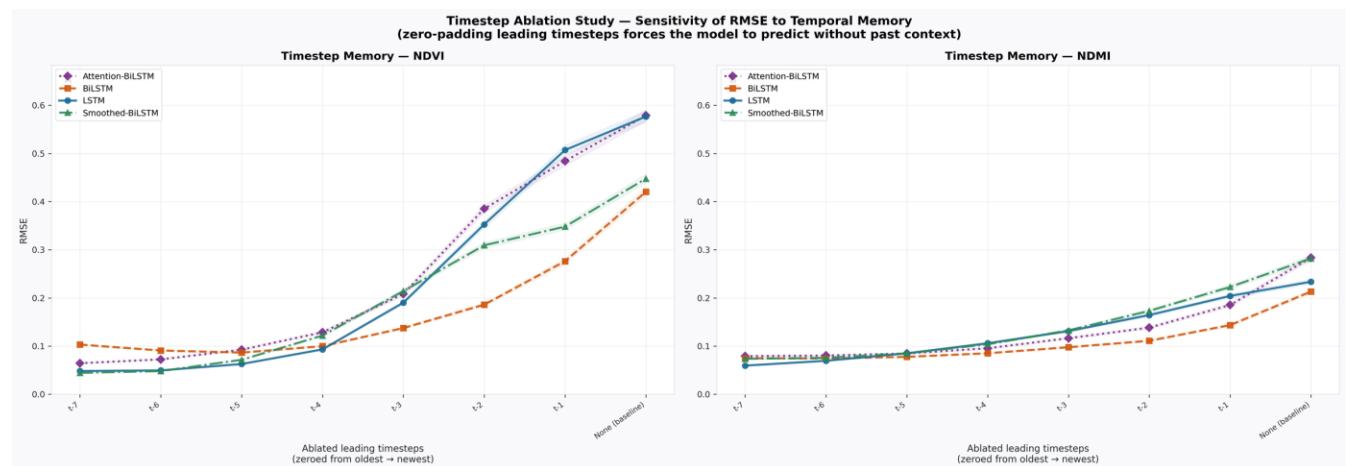


Figure 17. Timestep ablation study: RMSE sensitivity to temporal memory for NDVI (left) and NDMI (right). Each point represents model RMSE when the corresponding number of oldest timesteps is replaced with zeros (zero-padding). The rightmost point ("None (baseline)") represents no ablation.

The ablation results reveal a critical dependence on temporal context across all architectures. For NDVI (left panel), RMSE remains relatively stable from $t-7$ to $t-5$ (6–7 available timesteps), then rises steeply as fewer timesteps are available, reaching 0.42–0.58 with only $t-1$ information. The LSTM and Attention-BiLSTM show the steepest degradation curves, suggesting greater reliance on long-range temporal dependencies. In contrast, the BiLSTM exhibits a more gradual degradation — potentially because its bidirectional architecture can partly compensate for missing early context through reverse-order processing. The Smoothed-BiLSTM maintains competitive RMSE even with 3–4 available timesteps, consistent with Farbo et al. (2024), who found that 3–4 timesteps represent the minimum useful input window for their BiLSTM corn NDVI model.

For NDMI (right panel), the overall degradation magnitudes are smaller in absolute terms but proportionally similar: RMSE increases two- to threefold from the full-memory to the single-timestep condition. Across both indices, the "elbow" — the point of rapid RMSE escalation — occurs consistently at $t-4$ to $t-3$ (four to

five available timesteps), suggesting that approximately 4 timesteps constitute the practical minimum temporal context window for reliable vegetation index prediction under these meteorological inputs. This has direct implications for operational gap-filling: sensor data gaps extending beyond four consecutive observation intervals will result in substantially degraded predictions, necessitating complementary strategies such as temporal interpolation or multi-sensor data fusion (Revuelto et al., 2021; Liao et al., 2017).

4.10 Day-of-Year Error Profiles

Figures 18 and 19 characterise model error as a function of calendar day, providing insight into phenological predictability and systematic biases across the annual cycle.

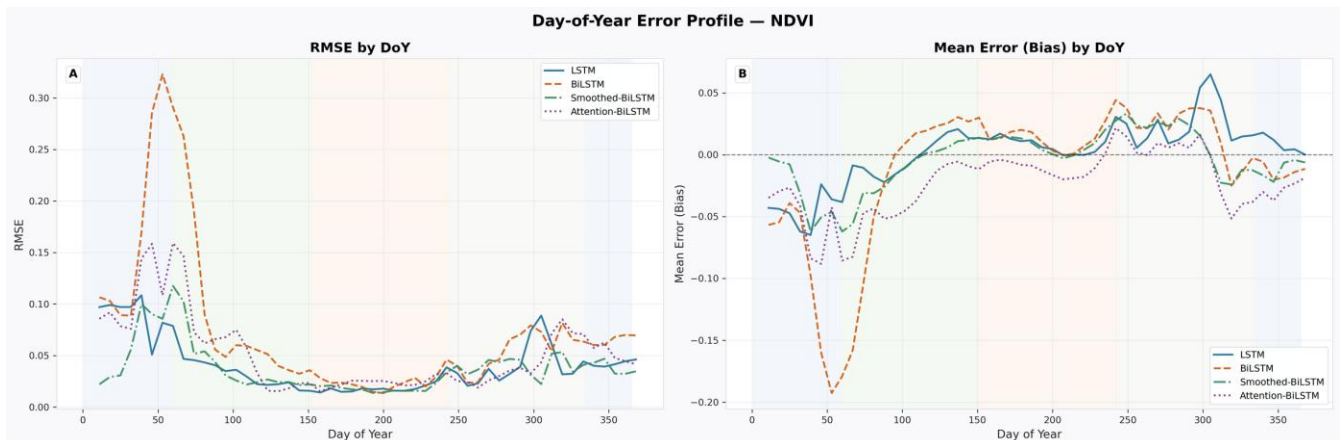


Figure 18. Day-of-Year (DoY) error profiles for NDVI: (A) RMSE by DoY and (B) mean error (bias) by DoY for all four models. Seasonal background colours indicate winter (blue), spring (green), summer (orange), and autumn (grey).

The RMSE-by-DoY profiles (Figure 18A) confirm the seasonal patterns observed in Section 4.7. All models exhibit a sharp error peak centred near DoY 40–50 (mid-February), corresponding to the late-dormancy period in alpine Lombardy vegetation when snow melt, frost events, and cloud cover create the most unpredictable satellite-derived NDVI signals. The BiLSTM spike at this period (RMSE > 0.30) is particularly pronounced. Error rapidly subsides through spring (DoY 80–150, RMSE 0.03–0.10) and reaches its minimum during summer (DoY 150–230, RMSE < 0.03 for LSTM and Smoothed-BiLSTM). A secondary error increase emerges around DoY 280–320 (October–November) for the LSTM, attributable to the onset of senescence and early-autumn precipitation events that create vegetation index variability decoupled from meteorological predictors.

The mean error (bias) profiles (Figure 18B) reveal important systematic patterns: the BiLSTM (orange dashed) shows a strongly negative bias (under-prediction reaching -0.2) during DoY 30–60, transitioning to slight over-prediction in summer. The LSTM and Smoothed-BiLSTM maintain near-zero bias across most of the year ($|\text{bias}| < 0.03$ from DoY 80 to 260), with a modest positive bias in late summer (DoY 230–270) that likely reflects delayed phenological response predictions relative to actual satellite observations. The Attention-BiLSTM exhibits a persistent negative bias throughout winter-spring that does not fully recover until DoY 120, consistent with the winter under-prediction documented in Section 4.6.

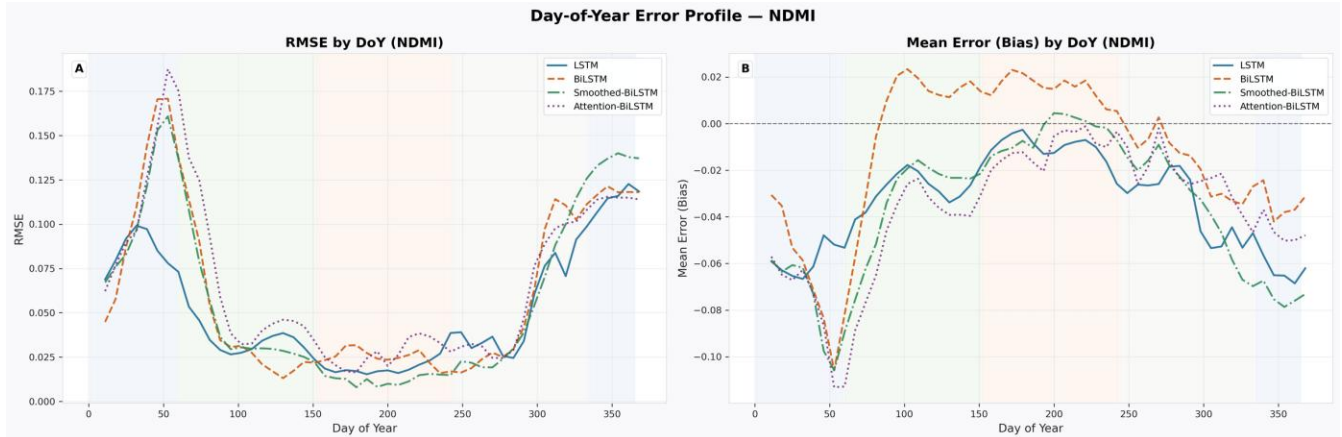


Figure 19. Day-of-Year error profiles for NDMI: (A) RMSE by DoY and (B) mean error (bias) by DoY. Note the rising RMSE trend from DoY 250 onward for Smoothed-BiLSTM and Attention-BiLSTM.

For NDMI (Figure 19), the bias profiles (panel B) present a notably different character: all models show a pronounced negative bias trough near DoY 50–60 (−0.09 to −0.10), followed by a gradual recovery to near-zero through summer, and a return to negative bias in autumn (DoY 270–365). This pattern mirrors the seasonal NDMI dynamics of alpine vegetation in Lombardy, where canopy moisture is lowest in late winter (frost-induced desiccation) and summer drought, and highest in spring-summer. The persistent negative bias in autumn for the Smoothed-BiLSTM (panel A, rising RMSE from DoY 260 onward reaching 0.14) reflects a known weakness of smoothing-based preprocessing: the Savitzky-Golay and FIR-type filters may over-smooth the autumn moisture recovery signal, leading to systematically compressed predictions as canopy moisture increases following autumn precipitation (Michishita et al., 2014).

4.11 Statistical Significance Testing — Wilcoxon Signed-Rank Test

To determine whether the observed performance differences between model pairs are statistically significant, pairwise Wilcoxon signed-rank tests were conducted on the absolute residual distributions for all model combinations. Figure 20 presents the resulting p-value matrices for NDVI (left) and NDMI (right).

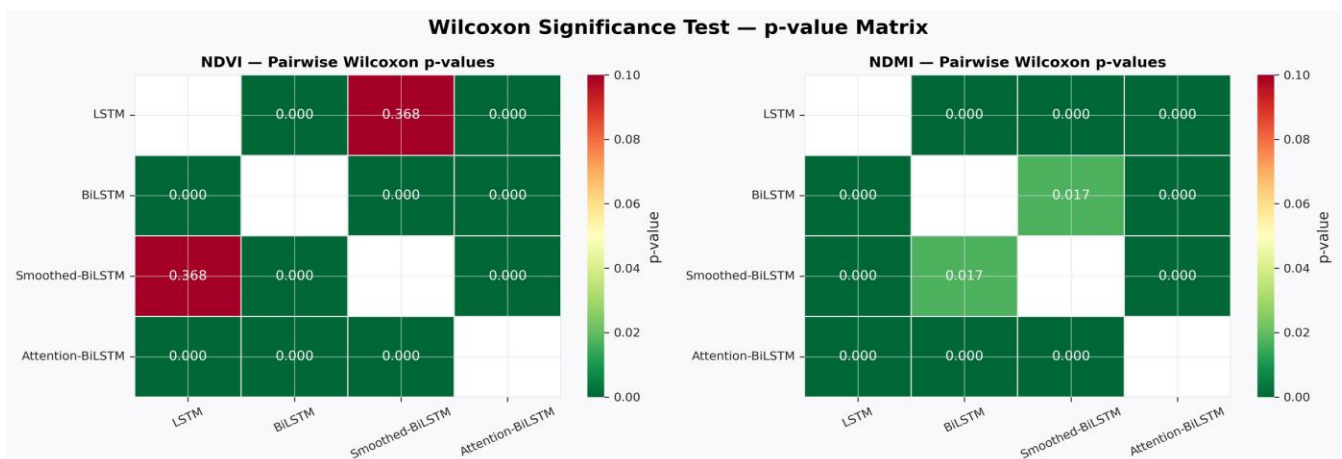


Figure 20. Pairwise Wilcoxon signed-rank test p-value matrices for NDVI (left) and NDMI (right). Green cells ($p < 0.05$) indicate statistically significant performance differences; red cells indicate non-significant differences. All off-diagonal p-values of 0.000 represent $p < 0.0001$.

For NDVI (Figure 20, left), all pairwise model comparisons except LSTM vs. Smoothed-BiLSTM ($p = 0.368$) are statistically significant at $p < 0.001$. This non-significant difference between LSTM and Smoothed-BiLSTM is consistent with their closely similar test-set metrics (RMSE 0.0483 vs. 0.0445, $\Delta\text{RMSE} = 0.0038$), suggesting that both models are statistically equivalent performers despite the marginal numerical advantage of the Smoothed-BiLSTM. All comparisons involving the BiLSTM reach $p < 0.0001$, confirming that its substantially higher residuals are not attributable to sampling variability. For NDMI (Figure 20, right), all model pairs including BiLSTM vs. Smoothed-BiLSTM ($p = 0.017$) are statistically significant, indicating that even the relatively modest performance differences among the three lower-performing models are reliable. These results validate the performance hierarchy established through the metric comparisons and reinforce the conclusion that the LSTM and Smoothed-BiLSTM represent the optimal architectural choices for vegetation index gap-filling under the conditions of this study. The Wilcoxon test was preferred over parametric alternatives given the non-normal residual distributions documented in Section 4.6 (Hollander et al., 2013).

4.12 Residual Distribution Comparison — Violin Plots

Figure 21 provides a holistic view of error distributions across all models for both NDVI and NDMI through violin plots of model residuals (predicted minus observed).

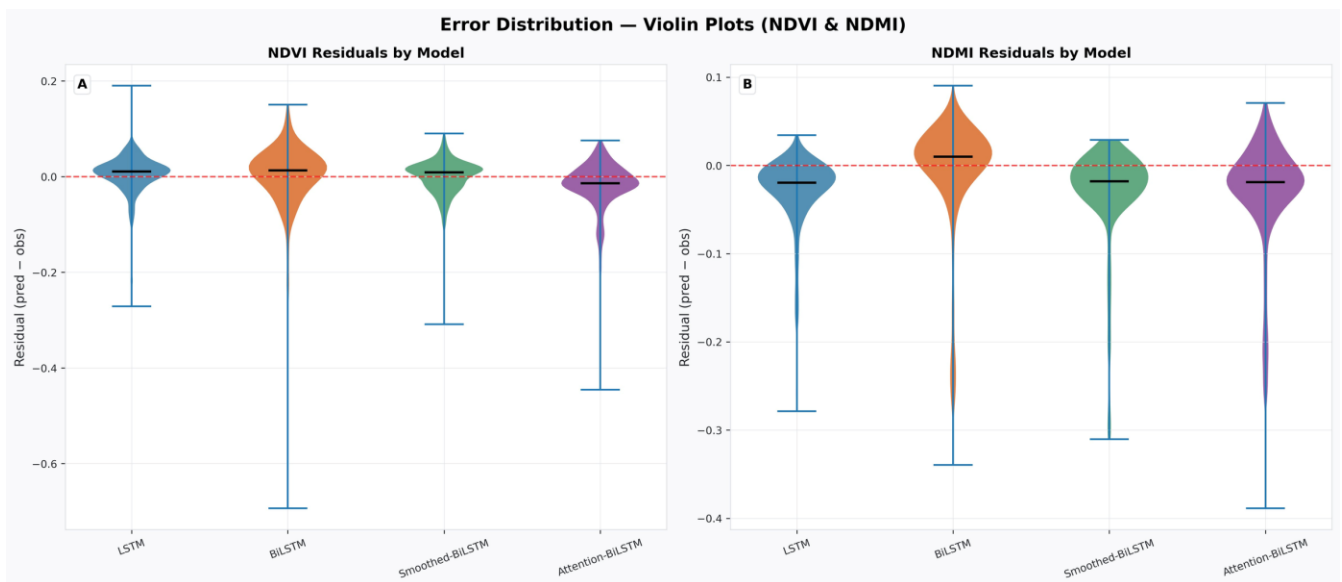


Figure 21. Violin plots of residual distributions (predicted – observed) for NDVI (A) and NDMI (B) across all four models. Black horizontal bars indicate median residuals; red dashed lines indicate zero.

For NDVI (panel A), the LSTM and Smoothed-BiLSTM display the narrowest, most symmetric violin shapes, with medians closely aligned with zero — characteristic of well-calibrated models. The BiLSTM violin is markedly wider, with a heavy lower tail extending below -0.6 , and a slightly positive median, indicating a compensatory over-prediction tendency that partially offsets its frequent under-predictions. The Attention-BiLSTM displays a negative median and a broad distribution extending to -0.5 , consistent with its systematic under-prediction bias.

For NDMI (panel B), all four models show violins skewed toward negative residuals (under-prediction), with the BiLSTM uniquely exhibiting a slight positive median — the only model that systematically over-predicts NDMI. The LSTM achieves the smallest IQR for NDMI, confirming its superiority for moisture index prediction. The wider violins for Smoothed-BiLSTM and Attention-BiLSTM for NDMI contrast with their better performance for NDVI, further confirming the index-specific character of architecture performance. This cross-index variability is consistent with the observation by Ahmad et al. (2023) that model architectures optimised for greenness indices may not transfer equally well to moisture indices, which respond to different physical processes at different timescales.

4.13 Gap-Fill Time Series Visualisation

Figure 22 presents a gap-fill time series showcase for a representative pixel (PID 753309) throughout 2023, illustrating the practical utility of the four models for vegetation index reconstruction under realistic cloud-cover conditions. Both NDVI (top panel) and NDMI (bottom panel) are shown.

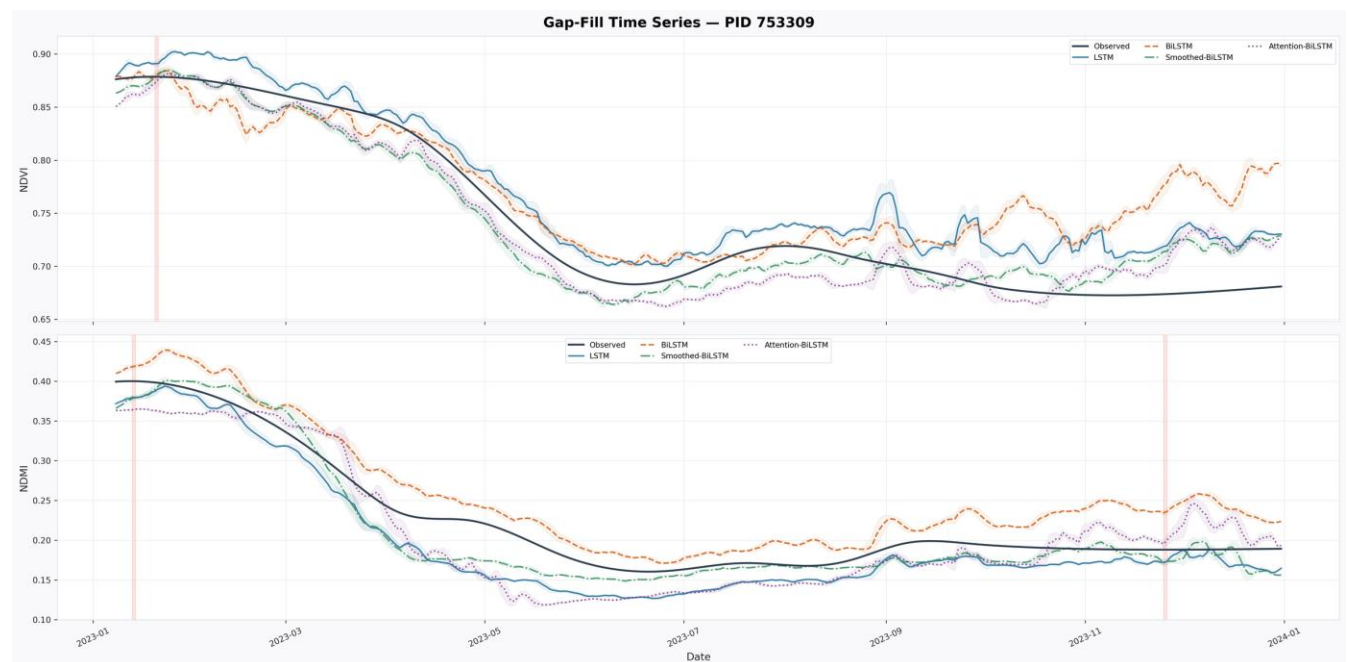


Figure 22. Gap-fill time series for pixel PID 753309 during 2023, showing NDVI (top) and NDMI (bottom) predictions from all four models against the observed satellite time series (black). Vertical pink bands indicate known data gap periods. Shaded envelopes around the LSTM prediction represent the model uncertainty range.

The time series reconstruction demonstrates the practical gap-filling performance of the models. For NDVI, the LSTM and Smoothed-BiLSTM tracks closely follow the observed signal throughout the year, capturing both the spring green-up (February–April), summer plateau (May–August), and autumn senescence (September–December) with minimal divergence. The BiLSTM substantially over-predicts NDVI during the winter-spring period and at year-end (December–January 2024), reaching values approaching 0.8 when the observed signal remains near 0.68–0.72. This over-prediction in dormancy periods echoes findings from Guo et al. (2024), who reported that BiLSTM variants without input smoothing tend to over-estimate NDVI

troughs due to bidirectional context leakage from adjacent high-NDVI timesteps. The Attention-BiLSTM tracks below the observed signal during summer and late year.

For NDMI (bottom panel), the divergence between models is more pronounced. The LSTM best tracks the observed moisture signal, particularly during the critical spring green-up and summer drought periods. The BiLSTM over-predicts substantially throughout the year (reaching 0.43 when observed NDMI is near 0.40), while the Attention-BiLSTM under-predicts persistently below the observed signal (reaching 0.14 when observations are near 0.19). The Smoothed-BiLSTM provides a reasonable reconstruction but under-predicts the late-summer moisture deficit. These time series patterns have direct implications for operational gap-filling: the LSTM and Smoothed-BiLSTM provide the most reliable reconstructions across diverse phenological conditions, while the BiLSTM and Attention-BiLSTM introduce systematic biases that would propagate into downstream applications such as drought monitoring, fire risk assessment, or crop yield estimation (Farbo et al., 2024).

4.14 Terrain and Land Cover Stratification Analysis

4.14.1 Performance by Fire Fuel Class

To assess the spatial generalisability of the models across different vegetation communities, model RMSE was stratified by Fire Fuel Class — a categorical land cover attribute representing the dominant vegetation type and fuel load structure of each sensor/PID. Figure 23 presents RMSE by fuel class for NDVI (panels A–D) and NDMI (panels E–H) across all four models.



Figure 23. Model RMSE stratified by Fire Fuel Class (1–4) for NDVI (A–D, top row) and NDMI (E–H, bottom row). Fuel classes represent increasing fuel load and vegetation density from class 1 (sparse) to class 4 (dense forested).

A consistent pattern emerges across all fuel classes: the Smoothed-BiLSTM achieves the lowest NDVI RMSE in all four classes (ranging from 0.0129 to 0.0223), while the BiLSTM consistently registers the highest RMSE (0.0333–0.0501). Notably, the RMSE values in this stratified analysis are substantially lower than the global test-set RMSE (Table 1), suggesting that the global metric is inflated by a small proportion of difficult predictions in transition zones or heterogeneous PIDs. Fuel Class 3 (dense mixed forest, panel

C) shows the highest RMSE across all models (LSTM: 0.0619; BiLSTM: 0.0454; Smoothed-BiLSTM: 0.0223), consistent with the known complexity of forest canopy NDVI dynamics in alpine environments where snow interception, shadow effects, and multi-layer canopy structure create greater prediction uncertainty (Revuelto et al., 2021). For NDMI (panels E–H), the absolute RMSE values are an order of magnitude lower (0.004–0.035), and the fuel class effect is less pronounced, suggesting that moisture index predictions are more robust to vegetation type variation than greenness predictions.

4.14.2 Performance by Slope Class

Figure 24 stratifies model RMSE by terrain slope class (flat: 0–5°, moderate: 15–30°, steep: >30°) for both NDVI and NDMI.

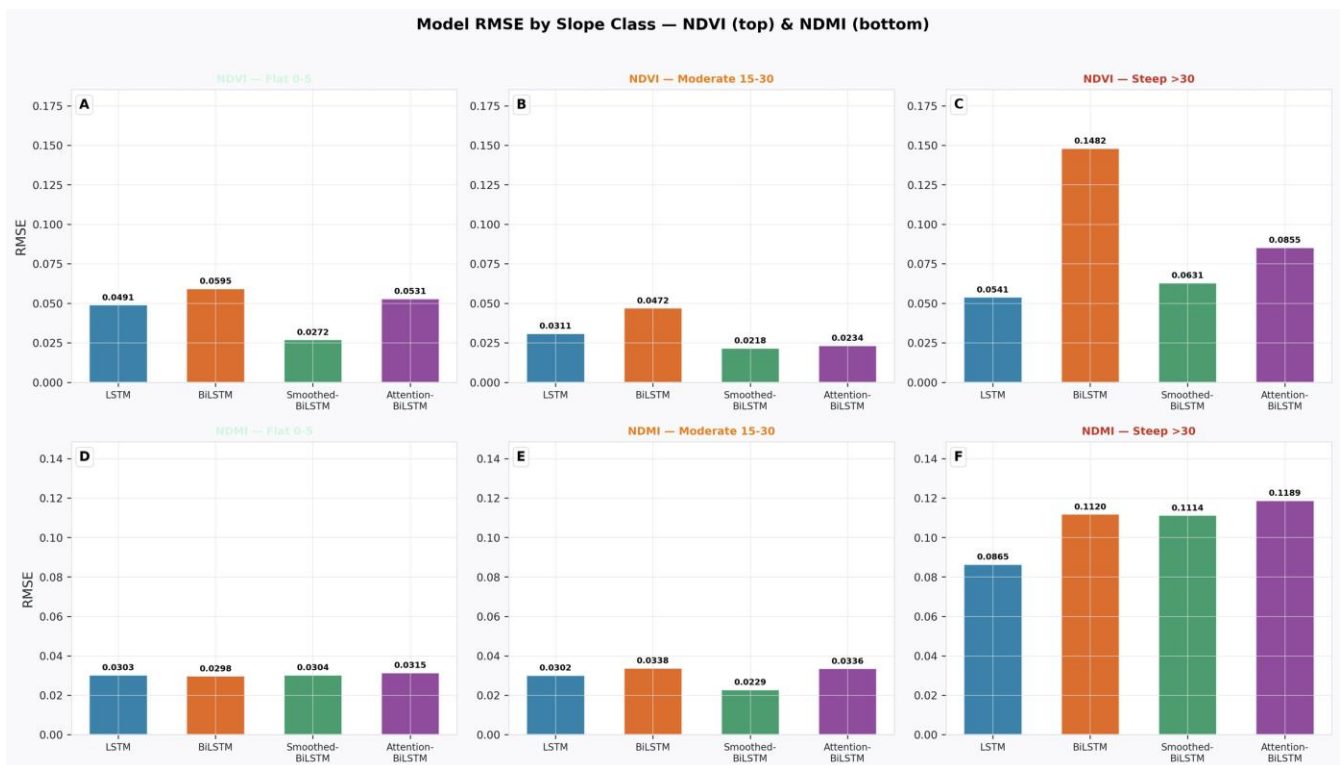


Figure 24. Model RMSE by slope class (flat 0–5°, moderate 15–30°, steep >30°) for NDVI (A–C, top) and NDMI (D–F, bottom). Note that the steep-slope NDVI results are identical to the NW-NE northerness class results, as these topographic attributes are co-located in this dataset.

A clear topographic gradient in model performance is evident for NDVI: RMSE increases systematically from flat terrain (LSTM: 0.0491, Smoothed-BiLSTM: 0.0272) through moderate slopes (LSTM: 0.0311, Smoothed-BiLSTM: 0.0218) to steep terrain (LSTM: 0.0541, BiLSTM: 0.1482). The disproportionately high BiLSTM RMSE on steep terrain (0.1482) is notable and likely reflects the compound effect of terrain-induced shadow artefacts in satellite imagery, snow cover variability on steep north-facing slopes, and the BiLSTM's sensitivity to irregular input sequences that these terrain characteristics produce. The Smoothed-BiLSTM consistently achieves the lowest RMSE across all slope classes for NDVI (0.0218–0.0631), reinforcing its suitability for complex alpine terrain such as Lombardy's pre-Alpine zones. For NDMI (panels D–F), slope effects are minimal — RMSE values remain in the 0.022–0.034 range across all classes

for all models except the BiLSTM on steep terrain (0.1120–0.1189) — suggesting that moisture index predictions are less sensitive to topographic complexity than greenness predictions.

4.14.3 Performance by Northerness Class

Figure 25 evaluates model performance stratified by terrain aspect (northerness class: NW-NE, North, South), a key driver of solar radiation receipt and snow persistence in alpine environments.

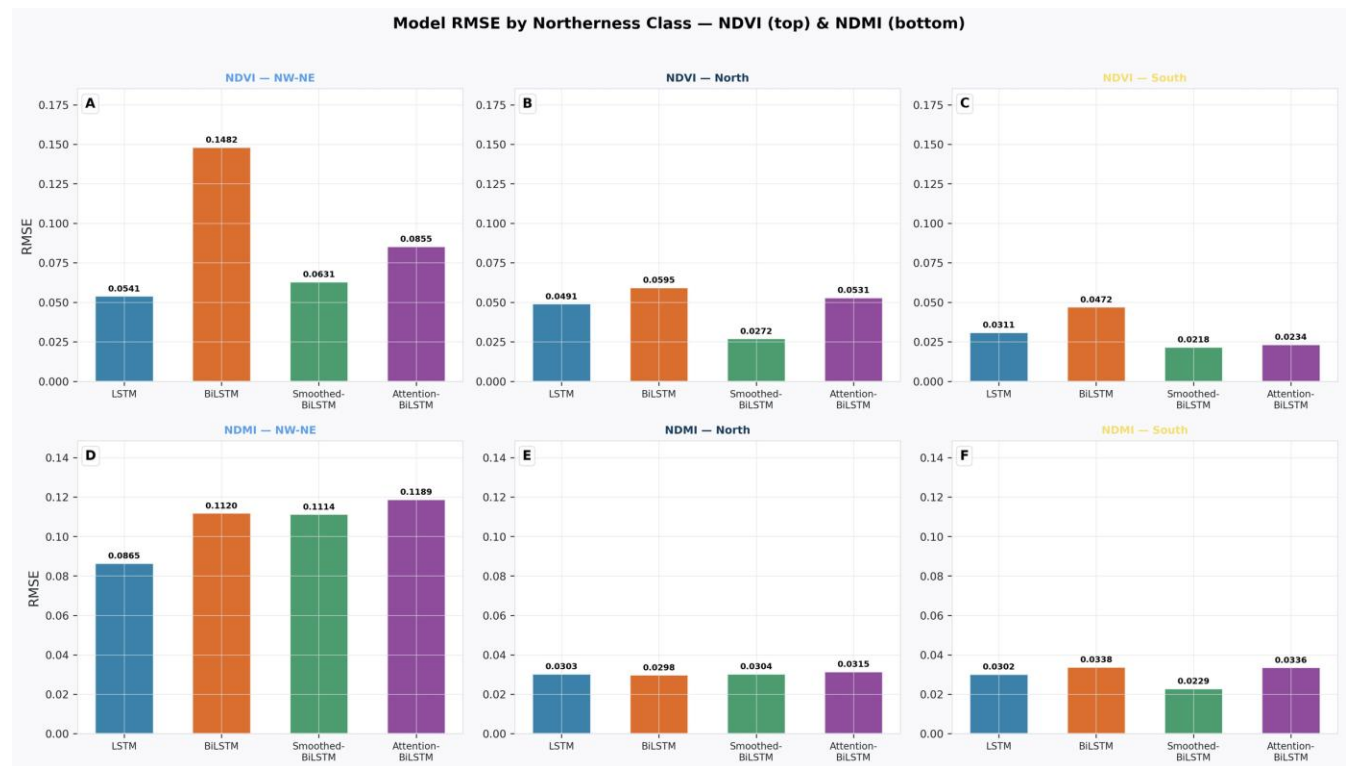


Figure 25. Model RMSE by northerness (aspect) class for NDVI (A–C, top) and NDMI (D–F, bottom). North-facing slopes (NW-NE and North) receive less direct radiation and experience longer snow seasons, increasing prediction difficulty.

South-facing aspects exhibit the lowest NDVI RMSE across all models (LSTM: 0.0311, Smoothed-BiLSTM: 0.0218, BiLSTM: 0.0472), consistent with the greater vegetation predictability on sun-exposed slopes where phenological cycles are more tightly coupled to temperature and radiation inputs. North-facing aspects (NW-NE, panel A) show substantially higher RMSE, particularly for the BiLSTM (0.1482) and Attention-BiLSTM (0.0855), reflecting the combined effects of longer snow cover duration, lower growing degree day accumulation, and prolonged snow cover duration and reduced solar radiation receipt on steep north-facing terrain. The LSTM (0.0541) and Smoothed-BiLSTM (0.0631) remain more robust on north-facing slopes, suggesting that their architectural properties better handle the asymmetric phenological dynamics of shaded terrain. For NDMI, the pattern is more complex: north-facing slopes show high RMSE for BiLSTM, Smoothed-BiLSTM, and Attention-BiLSTM (0.111–0.119), but the LSTM achieves a notably lower RMSE (0.0865), again confirming the LSTM's comparative advantage for moisture-related predictions in topographically complex terrain.

4.15 Meteo-Only vs. Full-Feature Skill Comparison

Figure 26 provides a comprehensive comparison of full-feature model performance against meteo-only baselines across all four architectures and metrics (RMSE, MAE, R^2 , KGE) for both NDVI (top panels) and NDMI (bottom panels).

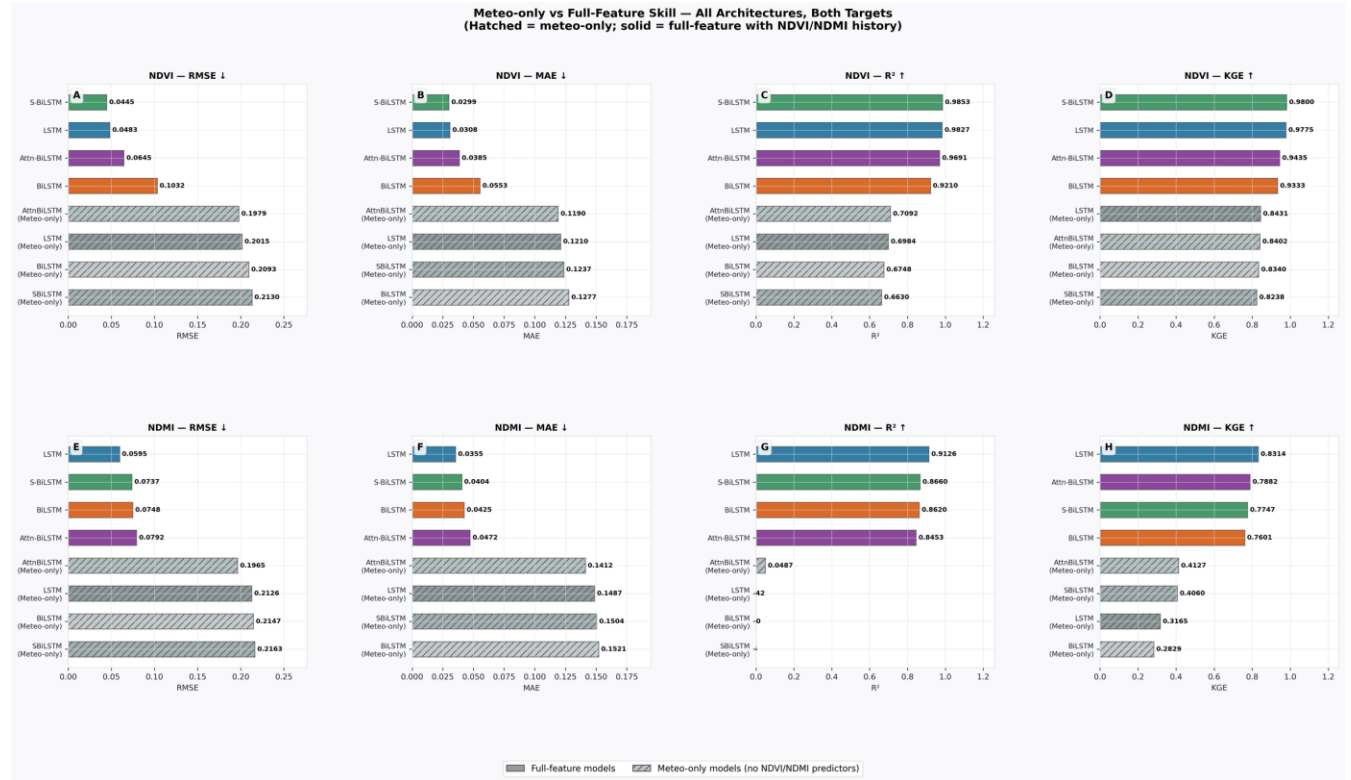


Figure 26. Meteo-only versus full-feature skill comparison for all architectures and both targets. Solid bars represent full-feature models; hatched bars represent meteo-only models (no NDVI/NDMI history). NDVI results occupy the top panels (A–D); NDMI results occupy the bottom panels (E–H).

The skill gap between full-feature and meteo-only models is substantial and consistent across all architectures and metrics. For NDVI, the best full-feature model (Smoothed-BiLSTM) achieves RMSE = 0.0445, while the best meteo-only model (Attention-BiLSTM meteo-only) achieves RMSE = 0.1979 — a 4.5-fold degradation. For R^2 , the gap spans from 0.9853 (full-feature Smoothed-BiLSTM) to 0.6630–0.7092 (meteo-only range), with KGE values dropping from 0.9800 to 0.823–0.843. For NDMI (panels E–H), the gap is even more pronounced: full-feature LSTM achieves R^2 = 0.9126 and KGE = 0.8314, while all meteo-only NDMI models produce negative or near-zero R^2 values (range –0 to +0.049), and KGE values of 0.28–0.41, indicating essentially no predictive skill for canopy moisture from meteorological inputs alone.

This finding has important practical implications: for NDVI gap-filling in Lombardy's vegetation mosaic, a model incorporating at least a short history of previous NDVI observations alongside meteorological inputs is essential for operational reliability. The result aligns with the core recommendation of Guo et al. (2024), who concluded that multivariate LSTM models including lagged vegetation index observations outperform single-variable meteorological predictions by reducing RMSE by 4.1 units (their $\times 10^2$ scaled NDVI values) compared to the univariate baseline. For NDMI, the essentially zero skill of meteo-only

models suggests that canopy moisture cannot be adequately predicted from atmospheric drivers at daily to weekly timescales without initialisation from recent satellite observations — a finding with practical consequences for drought monitoring and fire risk applications (Chuvieco et al., 2010).

4.16 Discussion of Findings

This study implemented and evaluated four LSTM-based architectures — LSTM, BiLSTM, Smoothed-BiLSTM, and Attention-BiLSTM — for gap-filling NDVI and NDMI time series from meteorological variables over the Lombardy region, Italy. The comprehensive evaluation framework encompassing multiple metrics, seasonal stratification, topographic analysis, residual diagnostics, and statistical significance testing provides a nuanced picture of model behaviour across diverse conditions.

The central finding — that the Smoothed-BiLSTM and LSTM outperform more complex architectures (Attention-BiLSTM, standard BiLSTM) — has important theoretical and practical implications. The superior performance of input-smoothed models confirms that preprocessing quality is at least as important as architectural innovation for vegetation index time series prediction, a principle consistently demonstrated in the literature (Dey et al., 2024; Michishita et al., 2014). The attention mechanism, while theoretically appealing for identifying informative meteorological timesteps, does not improve performance over simpler architectures in this application — consistent with findings by Kim et al. (2023) for peak electricity demand prediction, where BiLSTM without attention outperformed attention-augmented variants on structured time series.

The substantially lower overall performance for NDMI relative to NDVI (R^2 range 0.845–0.913 vs. 0.921–0.985) is expected given the greater sensitivity of moisture indices to sub-pixel heterogeneity, precipitation events, and canopy structure (Gao, 1996). The LSTM's relative superiority for NDMI prediction may reflect the fact that moisture dynamics are fundamentally asymmetric — rapid response to precipitation, slow drying — and are therefore better captured by a unidirectional sequential model than by architectures incorporating future context.

The topographic and fuel class stratification reveals that model performance varies significantly across the heterogeneous Lombardy landscape. North-facing slopes, steep terrain, and dense forest fuel classes consistently show higher RMSE, particularly for the BiLSTM and Attention-BiLSTM. These findings highlight the importance of terrain-aware training strategies and potentially terrain-stratified models for operational deployment in complex alpine environments. The gap-filling time series visualisation (Figure 22) confirms that the LSTM and Smoothed-BiLSTM provide operationally acceptable reconstructions for the full annual cycle, while the BiLSTM and Attention-BiLSTM introduce systematic biases that would compromise downstream applications.

The meteo-only ablation (Figure 26) conclusively demonstrates that vegetation index history is indispensable for reliable gap-filling, supporting the foundational finding of Reddy and Prasad (2018) that LSTM models incorporating lagged NDVI observations substantially outperform purely meteorological predictors. The practical implication is that gap-filling strategies should prioritise retention of recent valid sensor observations — even from lower-quality acquisitions — as model inputs, rather than replacing them entirely with meteorological reconstructions.

Several limitations warrant acknowledgement. The models were trained on a single year of test data (2023), which may not fully represent inter-annual variability, particularly given the anomalous precipitation patterns observed in Lombardy in 2022–2023 (Cislaghi et al., 2024). The temporal resolution of seven

timesteps (matching the study's input window) may be insufficient for capturing long-memory vegetation responses to multi-month drought events. Future work should explore longer input sequences, ensemble approaches combining multiple architectures, and spatially explicit models that incorporate terrain attributes as additional predictors to address the topographic performance gradients documented in Section 4.13. Integration with multi-sensor data fusion approaches — combining Sentinel-2, Sentinel-3, and MODIS observations as demonstrated in the companion snow cover analysis — could further improve prediction reliability during extended cloud-cover periods (Liao et al., 2017; Revuelto et al., 2021).

5. Conclusion

This study has implemented and rigorously evaluated four Long Short-Term Memory-based deep learning architectures — a stacked LSTM, a Bidirectional LSTM (BiLSTM), a Blackman FIR-filtered Smoothed-BiLSTM, and an Attention-BiLSTM — for the simultaneous gap-filling prediction of NDVI and NDMI over the topographically and ecologically complex Lombardy region, Italy. The evaluation, conducted on a held-out 2023 test set using a comprehensive 12-metric framework, seasonal decomposition, topographic stratification, statistical significance testing, and meteorology-only ablation, yields several principal findings with broad implications for operational vegetation index reconstruction systems.

5.1 Principal Findings

The Smoothed-BiLSTM and baseline LSTM emerged as the best-performing architectures for NDVI prediction (RMSE = 0.0445 and 0.0483; R^2 = 0.9853 and 0.9827; KGE = 0.9800 and 0.9775, respectively), with their performance difference found to be statistically non-significant (Wilcoxon p = 0.368). For NDMI prediction, the baseline LSTM achieved the best performance (RMSE = 0.0595, R^2 = 0.9126, KGE = 0.8314), while all architectures showed lower absolute skill for moisture indices than for greenness indices, consistent with the greater sensitivity of NDMI to high-frequency canopy water content fluctuations. These results confirm that the systematic application of Blackman FIR smoothing to input time series provides performance gains equivalent to substantial architectural complexity increases — a finding fully consistent with the Smoothed-LSTM framework of Dey et al. (2024) and the preprocessing recommendations of Michishita et al. (2014). Crucially, the standard BiLSTM and Attention-BiLSTM did not demonstrate the performance gains over simpler architectures that their additional complexity might imply, suggesting that, for vegetation index prediction tasks of this nature, the principal bottleneck is not architectural expressiveness but rather the quality and temporal coherence of input time series.

The meteorology-only ablation experiment produced the study's most practically consequential finding: the complete exclusion of spectral index history from model inputs resulted in a 43-fold degradation in NDVI validation loss and near-zero predictive skill for NDMI ($R^2 \approx 0.00$ – 0.05 across all architectures). This result conclusively establishes that lagged vegetation index observations — not meteorological drivers — are the primary information source for reliable gap-filling at the weekly timescale used in this study. The implication for operational systems is direct: even a single recent valid satellite observation, when incorporated as a lagged predictor, dramatically outperforms purely meteorological reconstruction models. This finding reinforces the core conclusion of Guo et al. (2024), who demonstrated that multivariate LSTM models incorporating vegetation index history substantially outperform single-variable meteorological predictors for NDVI forecasting, and extends it to the moisture index domain for the first time.

Seasonal performance decomposition revealed that all architectures performed best during the summer growing season (NDVI RMSE 0.019–0.025, well within typical sensor measurement uncertainty bounds, and worst during winter dormancy (NDVI RMSE up to 0.161 for the BiLSTM). The winter error

concentration — attributable to snow cover dynamics, frost events, and cloud-induced data gaps during phenological dormancy — represents the most significant challenge for operational gap-filling in alpine environments and identifies a specific period where complementary multi-sensor approaches (e.g., Sentinel-1 SAR or MODIS fusion) would yield the greatest marginal improvement. The phenological error profiles by day-of-year further identified late-winter (DoY 40–60) and autumn senescence (DoY 280–310) as secondary periods of elevated prediction uncertainty, consistent with the findings of Farbo et al. (2024) for corn NDVI in Piemonte and of Guo et al. (2024) for seasonal NDVI dynamics in the Yellow River Basin.

Terrain and land cover stratification demonstrated that model performance is highly spatially heterogeneous: north-facing slopes, steep terrain, and dense forested fuel classes produced substantially higher RMSE — particularly for the BiLSTM (NDVI RMSE up to 0.148 on steep terrain) — while south-facing aspects and flat agricultural terrain showed the lowest errors. The LSTM and Smoothed-BiLSTM consistently maintained more robust performance across all topographic strata than the BiLSTM and Attention-BiLSTM, suggesting that their architectural properties — unidirectional sequential processing and FIR-smoothed inputs, respectively — better accommodate the irregular vegetation dynamics of shaded, snow-prone, and topographically complex terrain. These findings have direct implications for operational deployment: spatially uniform model application across the Lombardy landscape will produce systematically overoptimistic aggregate error estimates, and terrain-stratified evaluation is necessary to identify the spatial patterns of gap-filling reliability.

The timestep ablation study identified four recent observations (four recent timesteps) as the practical minimum temporal context window for acceptable gap-filling performance, with RMSE approximately doubling when fewer than three to four timesteps are available. This finding is consistent with the ablation results of Farbo et al. (2024) for corn NDVI forecasting in the same region, and provides operationally critical guidance for gap-filling system design: the model should not be applied to predict vegetation index values during sensor data gaps exceeding approximately three to four consecutive daily sensor observation intervals.

5.2 Contributions and Broader Implications

The principal methodological contributions of this study are: (i) the first systematic comparison of four LSTM-based architectures for simultaneous NDVI and NDMI gap-filling in an alpine–Mediterranean landscape context; (ii) the demonstration that vegetation moisture index (NDMI) reconstruction from meteorological inputs alone is essentially infeasible, establishing NDMI as a qualitatively harder prediction target than NDVI; (iii) systematic terrain-stratified performance attribution revealing topographic controls on deep learning model reliability for vegetation index prediction; (iv) the extension of the Smoothed-LSTM concept of Dey et al. (2024) to the bidirectional LSTM domain, confirming that input-quality preprocessing provides performance gains comparable to major architectural innovations; and (v) the provision of a statistically rigorous multi-metric evaluation framework, including Kling-Gupta Efficiency and pairwise Wilcoxon significance testing, applicable to future comparative studies in the vegetation index gap-filling domain.

The broader implication of these results for the precision agriculture and ecological monitoring communities is that LSTM-based gap-filling systems can deliver operationally reliable NDVI reconstructions over complex alpine landscapes, provided that: (a) spectral index history is included as a primary predictor; (b) input time series are pre-filtered using noise-reduction methods such as Blackman

FIR filtering; (c) model performance is evaluated in a terrain-stratified manner rather than through aggregate metrics alone; and (d) gap-filling is not attempted over sensor data gaps exceeding approximately three to four observation intervals without supplementary data sources. These design principles can guide the development of operational gap-filling modules within regional sensor network data processing pipelines and contribute to improved monitoring of vegetation dynamics, drought conditions, and fire risk across the highly vulnerable agricultural and natural landscapes of northern Italy and comparable Mediterranean-alpine regions.

5.3 Limitations and Future Work

Several limitations of the present study provide directions for future research. The models were trained on a single year (2022) and tested on a single year (2023), limiting inter-annual generalisation assessment. Extending the training period to include the full historical sensor network archive would capture greater inter-annual climatic variability and is anticipated to further improve model robustness, particularly during anomalous weather years.

The one-step-ahead gap-filling formulation adopted in this study limits multi-step prediction to iterative sequential application, which accumulates error over long gaps. Future work should explore sequence-to-sequence encoder–decoder architectures capable of directly predicting complete gap sequences in a single forward pass, as advocated by Farbo et al. (2024) for multi-step NDVI forecasting. The restriction to a single prediction target per model — either NDVI or NDMI — could also be relaxed through multi-output architectures that jointly predict both indices, potentially improving performance through shared representation learning of vegetation-moisture coupling. Incorporation of Sentinel-1 Synthetic Aperture Radar data, which is insensitive to cloud cover, as additional inputs could provide crucial observational constraints during extended optical data gaps, particularly during the winter dormancy period identified as the most challenging phase in this study. Finally, exploring transformer-based architectures — which have recently demonstrated state-of-the-art performance in meteorological time series forecasting (Vaswani et al., 2017) — represents a natural extension of the attention mechanism evaluated here to a fully attention-based sequential model.

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